



Communication and Tracking System for a Stratospheric Balloon

Module : ITRI671

Denzel Verster

 **orcid.org/0009-0002-8074-6007**

Student number: 45036349

Degree and Year: *BSc Honours* in Computer Science and Information Technology 2026

Supervisor: Prof. Gunther Drevin

Co-supervisor: Mr. J. Heystek Grobler

Co-supervisor: Dr. Rudi Serfontein

Co-supervisor: Dr. Monica Barnard

Co-supervisor: Prof. Du Toit Strauss

Co-supervisor: Mr. Nathan Morgenrood

Co-supervisor: Mr. Bernard Swanepoel

Campus: Potchefstroom

Date: 22 June 2026

Phase: Empirical Study

Contents

List of Abbreviations	vii
1 Introduction	1
1.1 Introduction	1
1.2 Concepts Central to the Study	1
1.2.1 Telemetry Systems in High-Altitude Platforms	1
1.2.2 UHF-Based Telemetry Communication Systems	2
1.2.3 GPS-based tracking systems	2
1.2.4 Overview of Related Literature	3
1.3 Problem Description and Objectives	4
1.3.1 Problem Description	4
1.3.2 Research Question	4
1.3.3 Aims and Objectives	4
1.4 Research Methodology	5
1.4.1 Research Paradigm and Methodology	5
1.4.2 Data Collection	6
1.4.3 Research Methods	7
1.4.4 Data Analysis	7
1.4.5 The Artefact Project	8
1.4.6 Project Management and Project Plan	8
1.4.7 Development Platform, Resources, and Environment	10
1.5 Ethical Considerations	11

1.5.1	Information Leakage	11
1.5.2	Spectrum and Regulatory Compliance	11
1.5.3	Flight Safety and Payload Recovery	11
1.6	Provisional Chapter Division	11
1.7	Conclusion	12
2	Research methodology	13
2.1	Introduction	13
2.2	Research paradigms	13
2.2.1	Positivism	13
2.2.2	Interpretivism	14
2.2.3	Critical theory	14
2.2.4	Pragmatism	14
2.3	Research methodology	15
2.3.1	Design science research (DSR)	15
2.3.2	Applicability of design science research	16
2.3.3	Research rigour and validity	17
2.4	Research methods	17
2.4.1	Applied research methods:	17
2.4.2	Research procedure	18
2.5	Research design	18
2.6	Conclusion	19
3	Literature review	20
3.1	Introduction	20
3.2	Related literature	20
3.2.1	Telemetry systems in high-altitude platforms	21
3.2.2	UHF-based telemetry communication systems	22
3.2.3	GPS-based tracking systems	23

3.2.4	Additional considerations for high-altitude tracking and communication systems	24
3.3	How the literature guides this study	27
3.4	Conclusion	28
4	Empirical study	29
4.1	Introduction	29
4.2	Artefact description	30
4.2.1	Purpose and scope	30
4.2.2	System architecture	30
4.2.3	System requirements	31
4.3	Description of the development process	31
4.3.1	Problem identification and objective definition	32
4.3.2	Design and development cycle 1	32
4.3.3	Design and development cycle 2	35
4.3.4	Ground station design and testing	36
4.4	Findings	38
4.4.1	GPS-based testing results	38
4.4.2	Communication link testing results	41
4.4.3	Summary of findings	41
4.5	Conclusion	41
A	Programs	45
A.1	Ground-based GPS acquisition and logging program	45
A.2	GPS data analysis program	48
B		53
B.1	Code of Conduct	53
B.2	Student Study Leader Agreement	55
B.3	Declaration of Integrity	57

B.4 Ethics Form	59
---------------------------	----

List of Figures

1.1	Design Science Research (DSR) process model adopted from Peffers et al. [23] . .	6
1.2	Project Gantt Chart	9
2.1	Overview of the design science research (DSR) lifecycle	16
2.2	Research design followed in the study.	19
3.1	High-altitude balloon payload architecture proposed by Wiyagi et al. [35].	26
4.1	Verified UART wiring configuration between the Raspberry Pi Zero and Upu- tronics GPS expansion board.	33
4.2	Example of successful GPS data acquisition, showing consecutive valid GPS fixes containing latitude, longitude, altitude, speed, track, and the number of satellites used.	34
4.3	Wiring configuration between the Raspberry Pi Zero and the Radiometrix NTX2 UHF transmitter, based on [30].	35
4.4	Completed Radiometrix NTX2 UHF transmitter artefact.	36
4.5	Wiring configuration between the Radiometrix NRX2 receiver and the Arduino Uno R3 at the ground station.	37
4.6	Terminal output produced by the GPS log analysis program.	39

List of Tables

4.1	Quantitative summary of GPS-based GPS positional consistency.	38
-----	---	----

List of Abbreviations

ADR	Adaptive Data Rate
DSR	Design Science Research
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
ISM	Industrial, Scientific and Medical Band
JSON	JavaScript Object Notation
LEO	Low Earth orbit
LoRaWAN	Long Range Wide Area Network
LPWAN	Low-Power Wide-Area Network
NBFM	Narrowband Frequency Modulation
UART	Universal Asynchronous Receiver-Transmitter
UHF	Ultra High Frequency
WAN	Wide-Area Network

Chapter 1

Introduction

1.1 Introduction

Reliable communication at extreme altitudes remains a significant challenge in modern remote sensing and aerospace systems [15, 16]. Stratospheric research platforms are gaining more attention as alternative platforms for Earth observation and remote sensing applications [20].

Among these platforms, stratospheric research balloons that operate in the upper atmosphere provide an effective platform for the collection of atmospheric data and experimental research [33]. These types of systems require long-range communication and tracking capabilities to enable telemetry transmission to a ground station and the recovery of the stratospheric balloon [3, 25, 33]. The challenge of establishing communication and accurate position tracking of a stratospheric balloon is maintaining long-range communication under strict power, weight, and budget constraints [3, 25]. Low-power UHF communication systems operating in the 433 MHz band are commonly used for long-range telemetry applications due to their energy efficiency [24, 25]. Their suitability for balloon-based platforms has also been demonstrated in prior stratospheric deployments [3].

This study focuses on the design of a tracking and communication system that allows for essential telemetry data, like location data, to be efficiently transmitted to a ground station. This chapter covers the fundamental concepts that are relevant to this study, which is followed by a problem description and research objectives. The research methodology is outlined with ethical considerations as well. Finally, a provisional chapter division structure is presented to guide the remainder of this research project.

1.2 Concepts Central to the Study

1.2.1 Telemetry Systems in High-Altitude Platforms

Telemetry systems are mainly used to collect, process, and transmit data gathered from remote platforms to a receiving station [25]. In the context of stratospheric balloon systems, telemetry plays a crucial role in enabling real-time monitoring during flight. The transmitted data typically

includes variables such as GPS location, altitude, temperature, and system status [25, 3]. This information is essential not only for tracking the stratospheric balloon but also to increase the chances of payload recovery.

A telemetry system generally consists of three main components: data collection, data processing, and data transmission [25, 33]. Data collection involves the gathering of data from onboard sensors, such as GPS location from a GPS module. This data is then processed and formatted into a structure that is suitable for transmission. Finally, the data is transmitted to a ground station using a suitable communication medium [35].

High altitude telemetry systems have limited bandwidth and power usage constraints [13, 19, 25]. Limited transmission rates require careful selection and prioritisation of data to ensure that only essential information is communicated. The power limitations also require efficient transmission strategies, such as reducing transmission frequency and optimising packet size. These constraints emphasise the importance of designing telemetry systems that balance data accuracy, transmission reliability, and energy efficiency, particularly in applications such as stratospheric balloon systems where resources are limited.

1.2.2 UHF-Based Telemetry Communication Systems

Ultra High Frequency (UHF) communication systems operating in the Industrial, Scientific and Medical (ISM) bands, such as 433 MHz, are widely used for long-range, low-power wireless communication [19, 24]. These systems are suitable for stratospheric balloon applications due to their reliability in achieving long-distance line-of-sight communication with relatively low power consumption [3, 24, 33]. Unlike higher frequency communication technologies, UHF signals experience less attenuation over long distances and are more tolerant to environmental factors, making them effective for high-altitude applications where infrastructure-based communication, such as cellular networks, is unavailable [10, 16, 24].

In the context of this study, UHF communication provides a suitable and practical solution for maintaining continuous telemetry transmission between the stratospheric balloon and the ground station. Its low power requirements and long range capabilities make it well-suited to the constraints of high-altitude systems, where energy efficiency and communication reliability are critical.

1.2.3 GPS-based tracking systems

Global Positioning System (GPS) technology enables position tracking by receiving signals from multiple satellites and computing receiver location using signal timing [6]. GPS-based tracking systems are essential for monitoring the geographical location and trajectory of a stratospheric balloon during flight, allowing operators to track movement in real time and support payload recovery operations [6, 13, 25].

A GPS module in a system typically receives continuous satellite signals and outputs positioning data in standard formats [6]. This data is then processed and integrated into the telemetry system for transmission to a ground station. The integration of GPS data with telemetry

communication enables continuous situational awareness, which is critical for the successful operation of high-altitude balloon systems [6, 13].

Although GPS provides reliable global positioning data, its effectiveness in high-altitude applications depends on several factors, such as signal availability, antenna performance, and system integration [6, 13, 34]. Additionally, the transmission of GPS data must be carefully managed to balance tracking accuracy with power consumption and communication constraints, particularly when operating over long distances with limited bandwidth.

In the context of this study, GPS data should also be available to the onboard stratospheric balloon system to enable capabilities like geofencing, which will limit the available travel area of the stratospheric balloon. Furthermore, redundancy mechanisms for tracking are important to improve the likelihood of payload recovery in the event of communication failure. These considerations highlight the importance of efficient integration between GPS tracking and the communication subsystem [13].

1.2.4 Overview of Related Literature

Existing literature indicates that stratospheric balloon platforms are increasingly used as cost-effective alternatives to satellites for atmospheric sensing, experimentation, and remote data collection [20, 33]. These systems depend heavily on reliable telemetry and tracking subsystems to maintain communication with a ground station and support payload recovery.

Previous studies have demonstrated the use of UHF and sub-GHz communication technologies for long-range, low-power telemetry applications [3, 24]. These technologies are generally preferred over higher-frequency alternatives due to favourable propagation characteristics and lower power requirements. However, many existing studies focus on communication range or link feasibility rather than complete system integration.

GPS-based tracking systems are widely used for remote platform positioning and recovery operations [6, 13]. While GPS location acquisition is well established, effective transmission of positioning data remains dependent on the reliability of the telemetry subsystem.

Research also suggests that conventional cellular communication systems are not ideal for high-altitude platforms because network coverage is optimised for terrestrial users rather than aerial systems [4]. This limits the practicality of cellular-only solutions for stratospheric balloon missions.

Although prior studies provide valuable insight into communication technologies, telemetry architectures, and tracking systems, less attention has been given to integrated low-power solutions specifically designed for stratospheric balloon applications. A need therefore exists for a system that combines efficient telemetry transmission, reliable tracking, and practical recovery support under high-altitude operating constraints.

This study contributes by designing, implementing, and evaluating an integrated low-power communication and tracking system for a stratospheric balloon. The proposed artefact combines GPS-based positioning, UHF telemetry transmission, and recovery-oriented system fea-

tures within a single platform intended for operation under realistic power, range, and environmental constraints.

A more comprehensive review of related literature is presented later in Chapter 3.

1.3 Problem Description and Objectives

1.3.1 Problem Description

Stratospheric research balloon systems require reliable and efficient tracking and communication with a ground station for essential telemetry and payload recovery. However, operating a stratospheric balloon at altitudes of between twenty and thirty kilometres introduces several technical challenges, including low power constraints, long communication distances, and the absence of communication infrastructure like cellular networks [3, 4, 10, 16]. These challenges make it difficult to maintain continuous tracking and telemetry transmission [19].

Existing communication and tracking solutions, such as cellular-based systems, are primarily designed for terrestrial coverage and user distribution on the ground. Cellular networks are primarily designed for ground coverage rather than high-altitude aerial platforms, resulting in limited applicability for stratospheric balloon communication systems [4]. In addition, studies on aerial cellular coverage indicate that signal geometry, antenna orientation, and line-of-sight constraints significantly reduce reliability at high altitudes, making cellular communication unsuitable for sustained telemetry at stratospheric operating heights [4].

Additionally, the limited data rates of narrowband UHF communication systems require careful optimisation of telemetry payload size and transmission intervals [19, 24].

Although GPS will provide for global positioning capability, the reliable and effective transmission of this data to ground stations remains a challenge [6, 13]. Redundancies for tracking and payload recovery will also have to be implemented [3, 13].

This clearly indicates the need for a system-level design that integrates GPS tracking, efficient telemetry management, and long-range communication in a manner that ensures efficient position tracking and energy efficiency under stratospheric conditions.

1.3.2 Research Question

How can a reliable and power-efficient communication and tracking system be built for a stratospheric balloon system that can operate at high altitudes?

1.3.3 Aims and Objectives

This study aims to achieve the following:

- To design a telemetry system capable of collecting and transmitting essential flight data

to a ground station

- To develop a GPS-based tracking system for real-time global position monitoring
- Implement a long-range, low-power communication system suitable for high-altitude environments
- Optimise telemetry data transmission for optimal efficiency under bandwidth and power constraints
- Establish methods of redundancies for communication and payload recovery
- To enable reliable tracking and recovery of the stratospheric balloon payload
- To evaluate the performance of the system through GPS-based and communication-link testing

These aims and objectives guide the design, implementation and evaluation of the proposed communication and tracking system. By addressing both the technical challenges of high-altitude communication and the practical requirements of reliable tracking and payload recovery, this study aims to develop a reliable and efficient solution that meets the constraints of stratospheric balloon applications. Achieving these objectives is expected to address the research question and assess the feasibility of a low-power, long-range communication and tracking system at high altitudes. The expected contribution of this study is a validated prototype and design framework for reliable low-power balloon telemetry systems.

1.4 Research Methodology

1.4.1 Research Paradigm and Methodology

This study is grounded in the Pragmatic paradigm and will follow a Design Science Research (DSR) methodology. Pragmatism assumes that the value of knowledge lies in its practical applicability and usefulness in solving real-world problems [7]. This is suitable for the current study because the performance of the tracking and communication system is influenced by real technical and environmental factors such as radio propagation, atmospheric conditions, hardware limitations, and power constraints.

DSR is an approach to problem-solving that enhances knowledge through the creation of artefacts. DSR contributes to scientific and technological understanding by developing solutions that address practical problems [22, 23].

The Design Science Research (DSR) process follows an iterative methodology consisting of problem identification, artefact design and development, demonstration, and evaluation cycles. The process model guides the structured development and assessment of the proposed system artefact. Figure 1.1 illustrates the DSR process model used in this study.

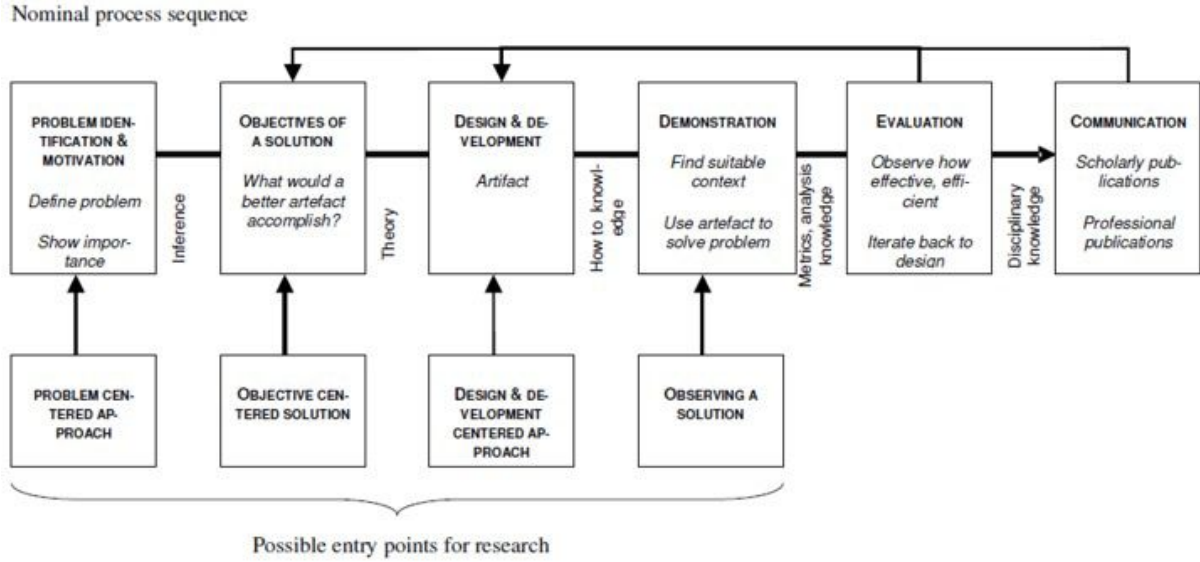


Figure 1.1: Design Science Research (DSR) process model adopted from Peffers et al. [23]

In this study, the artefact produced through the DSR process is a communication and tracking system for stratospheric balloon applications. This research methodology is applicable as this study aims to design and evaluate a communication and tracking system for a stratospheric balloon. Multiple stratospheric balloon launches will take place to test and evaluate the functioning, effectiveness, and efficiency of a communication and tracking system.

A quantitative research approach is selected, as the performance of the system will be evaluated using measurable metrics such as communication range, signal strength, packet transmission success rate, and GPS accuracy. This allows for objective assessment and comparison against defined performance criteria.

1.4.2 Data Collection

Data will be collected during the testing and evaluation of the communication and tracking system through a combination of GPS-based testing and communication-link testing.

The primary data collected will include:

- GPS coordinates (latitude, longitude, altitude)
- Signal strength indicators (RSSI)
- Packet transmission success and failure rates
- Transmission intervals and timing data
- System status data (e.g. power levels and system uptime)

Data will be collected using an onboard telemetry system consisting of a GPS module, single-board computer, and UHF transmitter. The transmitted telemetry data will be received by a ground station equipped with a UHF receiver and logged using data logging software.

Additional data may be collected through controlled ground testing to evaluate communication range and reliability under line-of-sight conditions [10]. All collected data will be stored in structured log files for further analysis.

1.4.3 Research Methods

The research methods applied in this study are aimed at evaluating the performance, reliability, and effectiveness of the proposed communication and tracking system under both controlled and real-world conditions. A combination of experimental, field-based and iterative development approaches will be used to ensure a comprehensive assessment of the system.

- Experimental testing - Measuring communication performance under controlled conditions
- Field testing - High-altitude balloon launches [29]
- Iterative prototyping - Improvements made based on results from each testing phase

These methods support the DSR approach by enabling the systematic design, testing, and improvement of the system. Experimental testing will provide for controlled performance evaluation, while field testing validates real-world applicability [23]. Iterative prototyping ensures continuous improvement of the system based on observed results, contributing to a robust and reliable final design.

1.4.4 Data Analysis

The collected data will be analysed to evaluate the performance, reliability, and efficiency of the communication and tracking system.

Key performance metrics will include:

- Maximum communication range
- Packet reception success rate
- Signal strength
- Transmission interval accuracy
- GPS accuracy and consistency

Data analysis will be performed using software tools capable of processing and visualising telemetry data. The collected data will be compared against expected performance benchmarks to determine whether the system meets the defined objectives.

Graphs and statistical summaries will be used to identify trends, such as signal degradation over distance and reliability under different conditions. The results of this analysis will be used to evaluate system performance and identify areas for improvement. The system will be evaluated against criteria such as reliable communication range, acceptable packet success rate, and consistent GPS accuracy during flight.

1.4.5 The Artefact Project

The artefact of this mini-dissertation is a communication and tracking system designed specifically for stratospheric balloon applications. The system integrates GPS-based tracking with a UHF telemetry communication link to enable reliable transmission of essential data to a ground station.

The system consists of the following components:

- A GPS module for acquiring real-time positional data
- A single-board computer for processing and formatting telemetry data
- A UHF transmitter and receiver pair for long-range communication

The purpose of the system is to provide reliable position tracking and telemetry transmission under high-altitude operation, limited power, bandwidth, and environmental constraints.

The development of the system will follow a structured lifecycle consisting of design, implementation, testing, and evaluation phases. Iterative improvements will be made based on testing results to optimise system performance and reliability.

1.4.6 Project Management and Project Plan

This project will be managed using an iterative approach aligned with the Design Science Research (DSR) methodology. The project will follow the core DSR phases, adapted for this study as follows:

- Problem identification and requirement analysis
- Definition of system objectives and specifications
- Design and development of the communication and tracking system
- Demonstration through GPS-based and communication-link testing
- Evaluation of system performance using defined metrics
- Refinement and optimisation based on evaluation results

These phases will be executed iteratively, allowing improvements to be made to the system after each testing and evaluation cycle. Milestones will be defined for each phase to track progress and ensure timely completion of the project.

The milestones include completion of system design, successful ground testing, initial flight testing, and final system evaluation.

Below, a Gantt chart Figure 1.2 is used to visually represent the project timeline, including task durations, dependencies, and key milestones.

Gantt Chart

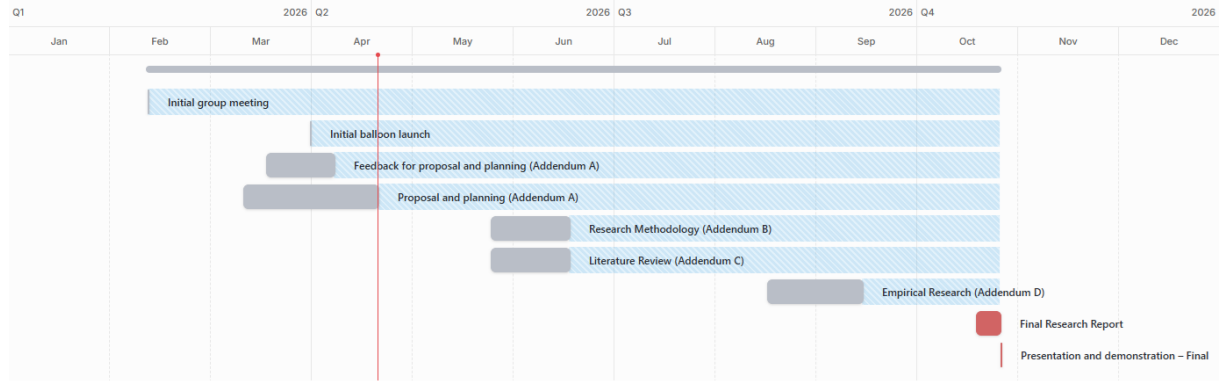


Figure 1.2: Project Gantt Chart

Integration with Group Project

This study forms part of a broader group project that involves three Honours students, where each student is tasked with developing a subsystem of a stratospheric balloon.

The project is divided among students as follows:

- This study focuses on the development and design of a tracking and communication subsystem, which is responsible for telemetry transmission and position tracking of the stratospheric balloon.
- A second project focuses on the design of the control system and onboard software environment, which manages system processes, payload release mechanisms and geofencing capabilities.
- The third project focuses on a sensor subsystem that is primarily responsible for collecting atmospheric data using a particulate sensor during the flight of the stratospheric balloon.

The communication and tracking subsystem will receive processed data from the control system and sensor subsystem, while integrating GPS data within the communication subsystem, and then package the information into telemetry data and transmit it to a ground station. This establishes a clear dependency where the system relies on accurate data input from other subsystems, while simultaneously enabling system-wide monitoring and recovery capabilities.

Although the overall project is developed collaboratively, this study maintains a clear focus on the communication and tracking subsystem. The design, implementation, and evaluation of this subsystem are conducted independently, ensuring that the research contribution remains distinct and aligned with the objectives of this study.

Scope, Limitations, and Risk Analysis

The scope of this study is limited to the design and evaluation of a communication and tracking system for a stratospheric balloon operating in the UHF frequency band [24, 10]. The system

will focus on transmitting essential telemetry data, including GPS location and system status, to a ground station.

Limitations of the study include constraints related to power consumption, hardware capabilities, and environmental factors such as atmospheric conditions. Additionally, the system will be evaluated under controlled test conditions, which may not fully represent all real-world scenarios.

Potential risks include communication signal loss, hardware failure after launch, and challenges in payload recovery. These risks will be mitigated through the implementation of redundancy mechanisms, careful system design, and pre-flight testing procedures.

1.4.7 Development Platform, Resources, and Environment

The development of the communication and tracking system will utilise a combination of hardware and software resources. The hardware platform will include a UPUTRONICS GPS module for location tracking [6], a RADIOMETRIX UHF transmitter and receiver operating in the 433 MHz band for communication [10, 24], and a Raspberry Pi single-board computer for data processing and system control [32].

Software development will be conducted using embedded programming environments suitable for single-board computer-based systems, primarily using Python for system control and data handling on the Raspberry Pi, which is widely used in low-power embedded and prototype telemetry systems [32]. Data analysis and testing will be supported by telemetry logging utilities and data visualisation tools such as Python-based plotting libraries.

The components consist of the following:

- UPUTRONICS GPS module
- RADIOMETRIX UHF transmitter
- RADIOMETRIX UHF receiver
- Raspberry Pi single-board computer
- Power supply system (battery-based)

The selected components are chosen based on their low power consumption, reliability, and suitability for high-altitude and long-range telemetry applications, as supported by existing LPWAN and stratospheric balloon studies [3, 24, 33]. The development environment enables fast prototyping and testing, enabling iterative improvement of the system design.

1.5 Ethical Considerations

1.5.1 Information Leakage

Telemetry systems used in remote sensing and high-altitude platforms may unintentionally expose operational information even when the transmitted data packet is encrypted. This is possible, as shown in [17], because information can still be deduced by an outside observer from transmission behaviour alone, even if the data being transmitted has been encrypted.

To mitigate leaking information, this system will only transmit essential data, such as GPS location, in time intervals of around 2 minutes.

1.5.2 Spectrum and Regulatory Compliance

The system operates in the 433 MHz ISM band, which is subject to regulatory constraints on transmission power and usage conditions [11, 12]. Therefore, the design of the telemetry system considers regulatory limitations associated with low-power telemetry systems to ensure compliant operation.

The use of sub-GHz ISM band communication is consistent with low-power wider area networking principles, where long-range connectivity is achieved under strict energy and spectrum constraints [24].

1.5.3 Flight Safety and Payload Recovery

Stratospheric balloon systems introduce operational risks related to uncontrolled descent, payload separation, and environmental interaction during recovery, as identified in high-altitude platform system studies [8, 33]. These risks must be managed to ensure the safety of people, property, and equipment.

To address these risks, controlled ground testing is performed before launching the stratospheric balloon, and recovery procedures are pre-planned based on expected flight trajectories [15]. Redundant tracking mechanisms are also considered to improve payload location accuracy in the event of communication failure [3, 33].

These measures ensure that system testing and deployment are conducted in a controlled and responsible manner, consistent with experimental aerospace safety practices.

1.6 Provisional Chapter Division

This mini-dissertation will contain the following chapters:

1. Introduction:

Chapter 1, the introduction to this research project, describes the context and important concepts surrounding the tracking and communication of a stratospheric research balloon system.

2. Research Methodology:

The research methodology chapter discusses the research methodology and research paradigm selected to evaluate and improve a tracking and communication system for a stratospheric balloon.

3. Literature Review:

This chapter discusses the related literature that is referenced throughout this research project. This will include articles related to the communication and tracking of a remote research platform that accounts for communication with a ground station for GPS telemetry.

4. Empirical Research:

This chapter focuses on the design, development, and testing of the proposed communication and tracking system. A design and development approach will be followed, where the system is implemented, tested, and refined through iterative cycles. The performance of the system will be evaluated based on the collected data and defined performance metrics.

5. Conclusion:

Finally, the conclusion chapter provides the final synthesis of the research project. The conclusion consists of the findings and results of the developed stratospheric research balloon system.

1.7 Conclusion

Chapter 1 discussed the research topic by outlining the context, challenges, and objectives associated with communication and tracking in stratospheric balloon systems. Key concepts of the system relevant to the study were discussed, along with an overview of related literature to establish the foundation for the research.

The problem was defined, a research question, aims, and measurable objectives were formulated to guide this mini-dissertation. The proposed research methodology, project plan, and system design were also explained, along with considerations related to ethics and project execution.

The next chapter will further elaborate on the DSR methodology, providing a discussion of the research paradigm, methods, and design that will be used to conduct this study.

Chapter 2

Research methodology

2.1 Introduction

This chapter entails the philosophical and methodological approaches adopted for the study. The chapter starts by examining research paradigms relevant to computing research and motivates why the Pragmatic paradigm is selected. Thereafter, the design science research (DSR) methodology is discussed in relation to the development of the proposed artefact. The chapter further explains the research methods, data collection procedures, and data analysis techniques used during the study.

The following content in this chapter consists of the paradigms applicable to this study, followed by an explanation of the selected research methodology and the procedures followed to ensure rigour and validity. This chapter also presents the research methods used in the study and concludes with a detailed discussion of the research design and overall structure of the study.

2.2 Research paradigms

Research paradigms provide the philosophical foundation that guides how knowledge is created, interpreted, and evaluated within a study [14]. In computing and information systems research, several paradigms are typically applied depending on the nature of the research problem and the objectives of the study [23]. This study considers several research paradigms to determine their applicability for the development and evaluation of the proposed communication and tracking system.

2.2.1 Positivism

Positivism is based on the assumption that reality exists independently of human interpretation and can therefore be objectively measured and analysed through observable data [14, 21]. Positivist studies usually make use of quantitative methods, experimentation, and statistical analysis to produce measurable and reliable findings [5, 28].

The positivist paradigm is partially applicable to this study because the proposed system will be evaluated using measurable performance metrics like communication range, signal strength, packet transmission success rate, and GPS accuracy [21, 28]. These measurements enable objective evaluation of the system under controlled and operational conditions [23, 28].

This study also focuses on the iterative development and refinement of a technological artefact intended to solve a practical engineering problem [22, 23]. The study, therefore, extends beyond pure observation and measurement, making the positivist paradigm not fully applicable [23].

2.2.2 Interpretivism

Interpretivism focuses on understanding human experiences, perceptions, and subjective meanings within a social context [5, 14]. Interpretivist studies generally make use of qualitative methods such as interviews, observations, and thematic analysis to understand how individuals interpret their experiences [5, 27].

Although interpretivism is valuable in studies involving human behaviour and social interaction, it is an unsuitable paradigm for this study [5, 28]. This study does not investigate subjective human experiences or perceptions, but rather focuses on the technical design and evaluation of a communication and tracking system based on measurable engineering performance criteria.

2.2.3 Critical theory

Critical theory is a paradigm concerned with identifying and addressing societal inequalities, power structures, and social transformation [5, 9]. The research conducted using this paradigm often focuses on challenging dominant systems and promoting social change [5].

Critical Theory is not applicable to this study because the research does not involve political, social, or ideological critique. Instead, the study focuses on the technical development and evaluation of a telemetry and tracking artefact for high-altitude balloon applications.

2.2.4 Pragmatism

Pragmatism focuses on practical problem solving and the usefulness of knowledge in real-world applications [5, 7]. Pragmatic research emphasises selecting methods and approaches that address the research problem the best, rather than adhering strictly to a single philosophical position [5, 7].

The Pragmatic paradigm is suitable for this study because the research aims to design and evaluate a practical engineering solution to the communication and tracking challenges associated with stratospheric balloon systems. The effectiveness of the proposed system depends on real-world operational factors like communication reliability, atmospheric conditions, signal propagation, hardware constraints, and power efficiency [23, 25, 33].

This study focuses on developing a functional and reliable system capable of operating effectively under high-altitude conditions. The Pragmatic paradigm supports the combination of engineering design, experimentation, iterative improvement, and quantitative evaluation required to achieve the objectives of this study.

Based on the characteristics of the study and the objectives of the research, the Pragmatic paradigm is considered the most appropriate philosophical foundation for this investigation [5, 7].

2.3 Research methodology

Research methodology refers to the systematic approach used to conduct research within the context of a specific research paradigm. The methodology guides how the study is structured, how knowledge is generated, and how the validity and reliability of the findings are ensured [5, 23].

2.3.1 Design science research (DSR)

This study follows a DSR methodology. DSR focuses on the creation, development, and evaluation of artefacts intended to solve identified real-world problems [22, 23].

DSR contributes to scientific and technological knowledge through the development of innovative artefacts and the evaluation of their effectiveness in addressing practical challenges [22, 23]. The methodology is commonly applied in computing and information systems research where technological systems or solutions are developed to address engineering or operational problems [2, 22].

The DSR process generally consists of the phases illustrated in Figure 2.1

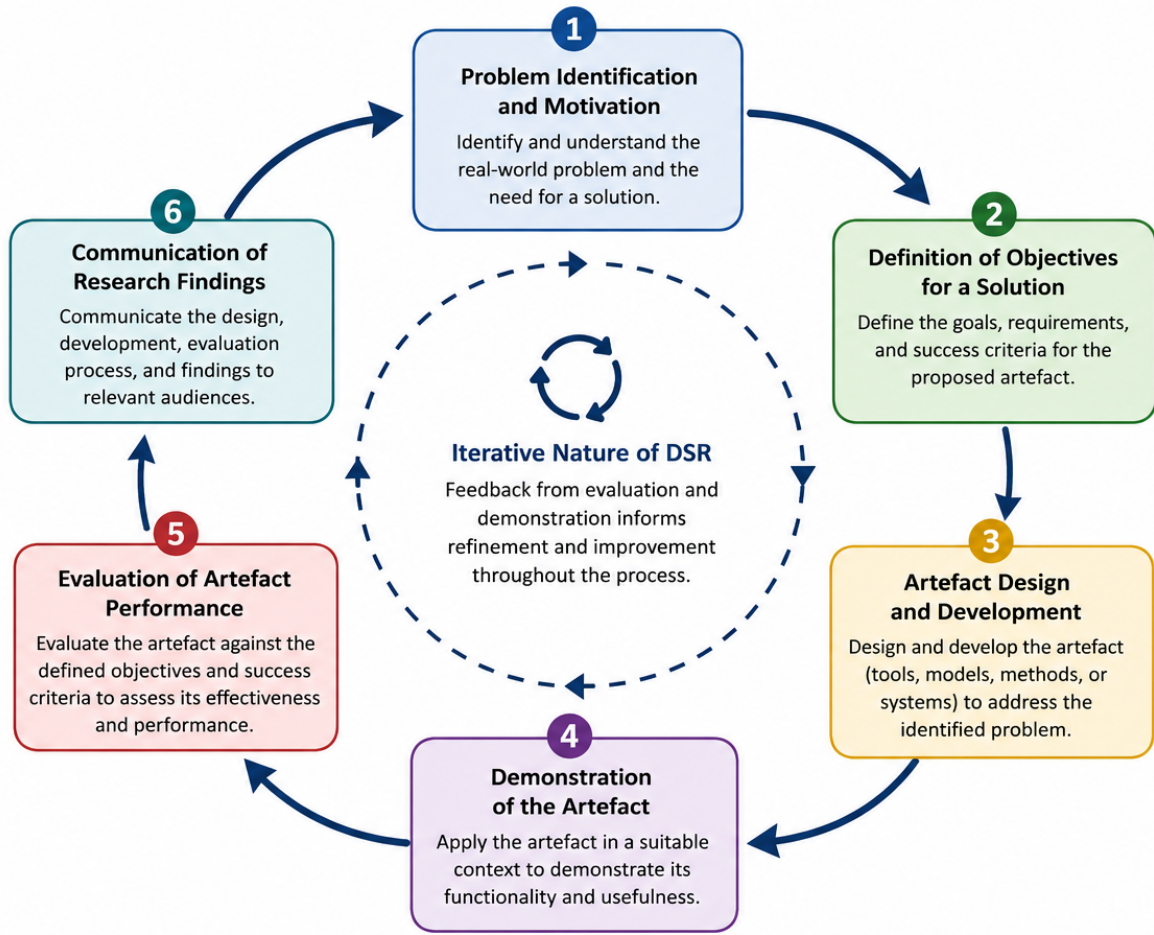


Figure 2.1: Overview of the design science research (DSR) lifecycle

The iterative nature of DSR enables continuous refinement and improvement of the artefact throughout the research process [2, 23].

2.3.2 Applicability of design science research

The DSR methodology is appropriate for this study as it focuses on the development of an artefact intended to solve a practical engineering problem. In this case, the artefact is a communication and tracking system for stratospheric balloon applications, designed to address challenges related to telemetry transmission and position tracking under high-altitude constraints [22, 23, 25, 35].

DSR is further suitable because the artefact will be iteratively designed, implemented, and evaluated through both controlled experiments and real-world high-altitude balloon testing [23, 29, 33]. The outcomes of each evaluation cycle are used to refine and improve system performance, aligning with the iterative nature of the DSR process [23].

2.3.3 Research rigour and validity

Research rigour refers to the systematic and transparent execution of research to ensure that findings are reliable and reproducible [5]. In this study, rigour is achieved through structured testing procedures, repeatable experiments, and consistent data collection methods. Rigour is further reinforced through the iterative development cycles of the DSR methodology, in which continuous evaluation and refinement of the system occur throughout the study [2, 23].

Reliability is supported through repeated experimental and field tests under controlled conditions [5]. Communication range, GPS accuracy, and telemetry performance are measured multiple times using standardised logging procedures to ensure consistency in results.

Validity is also ensured through clearly defined performance metrics aligned with the study objectives, including signal strength, packet success rate, transmission consistency, and GPS accuracy [5]. These metrics ensure that system evaluation reflects actual operational performance requirements [5]. External validity is strengthened through high-altitude balloon testing, which exposes the system to real environmental and atmospheric conditions relevant to its intended application [29, 33, 35].

2.4 Research methods

The study applies a combination of experimental, field-based, and analytical research methods to evaluate the proposed communication and tracking system. These methods support the iterative nature of the DSR methodology by enabling continuous refinement of the artefact based on measurable performance outcomes. The selected methods focus primarily on evaluating communication performance, telemetry reliability, positioning accuracy, and system robustness under realistic operating conditions.

2.4.1 Applied research methods:

- Experimental testing:

Experimental testing involves evaluating the communication and tracking system under controlled conditions to measure performance metrics such as communication range, signal strength, packet transmission success rate, and GPS accuracy [15, 25]. This allows objective assessment of system functionality under repeatable conditions.

- Field testing:

Field testing involves high-altitude balloon launches to evaluate system performance under real operational conditions. These tests assess telemetry reliability, GPS tracking continuity, and communication stability in uncontrolled atmospheric environments [29, 33, 35]. Field testing is necessary because environmental conditions such as altitude, atmospheric variation, and distance significantly affect communication performance.

- Iterative prototyping:

Iterative prototyping requires continuously refining the communication and tracking system based on results obtained from both controlled and field testing. Each iteration improves system performance, reliability, and efficiency in alignment with DSR principles [23].

- Quantitative data analysis:

Quantitative data analysis involves evaluating telemetry data using measurable performance metrics such as communication range, signal strength, packet transmission success rate, GPS accuracy, and transmission interval consistency [5]. Statistical summaries and graphical analysis are used to assess system performance objectively.

2.4.2 Research procedure

The research procedure follows an iterative DSR process consisting of requirements analysis, literature review, system design, implementation, controlled experimental testing, field testing, quantitative data analysis, and iterative refinement of the artefact [23].

The study begins with identifying communication and tracking requirements for high-altitude balloon systems through literature analysis [25, 33, 35]. Based on these requirements, the communication and tracking system is designed and implemented using selected hardware and software components. Controlled testing is then conducted to evaluate baseline system performance under repeatable conditions before progressing to real-world field testing during high-altitude balloon launches.

Telemetry and tracking data collected during testing are analysed quantitatively using various performance metrics such as communication range, signal strength, packet transmission success rate, and GPS accuracy. The results gathered from testing phases are used to refine and improve the system iteratively until acceptable operational performance is achieved.

2.5 Research design

The research design follows an iterative DSR structure consisting of problem identification, system design, implementation, testing, quantitative evaluation, and artefact refinement. Controlled experimental testing is conducted before real-world field testing through high-altitude balloon launches. Results obtained during evaluation are used to refine the system iteratively until the required performance objectives are achieved.

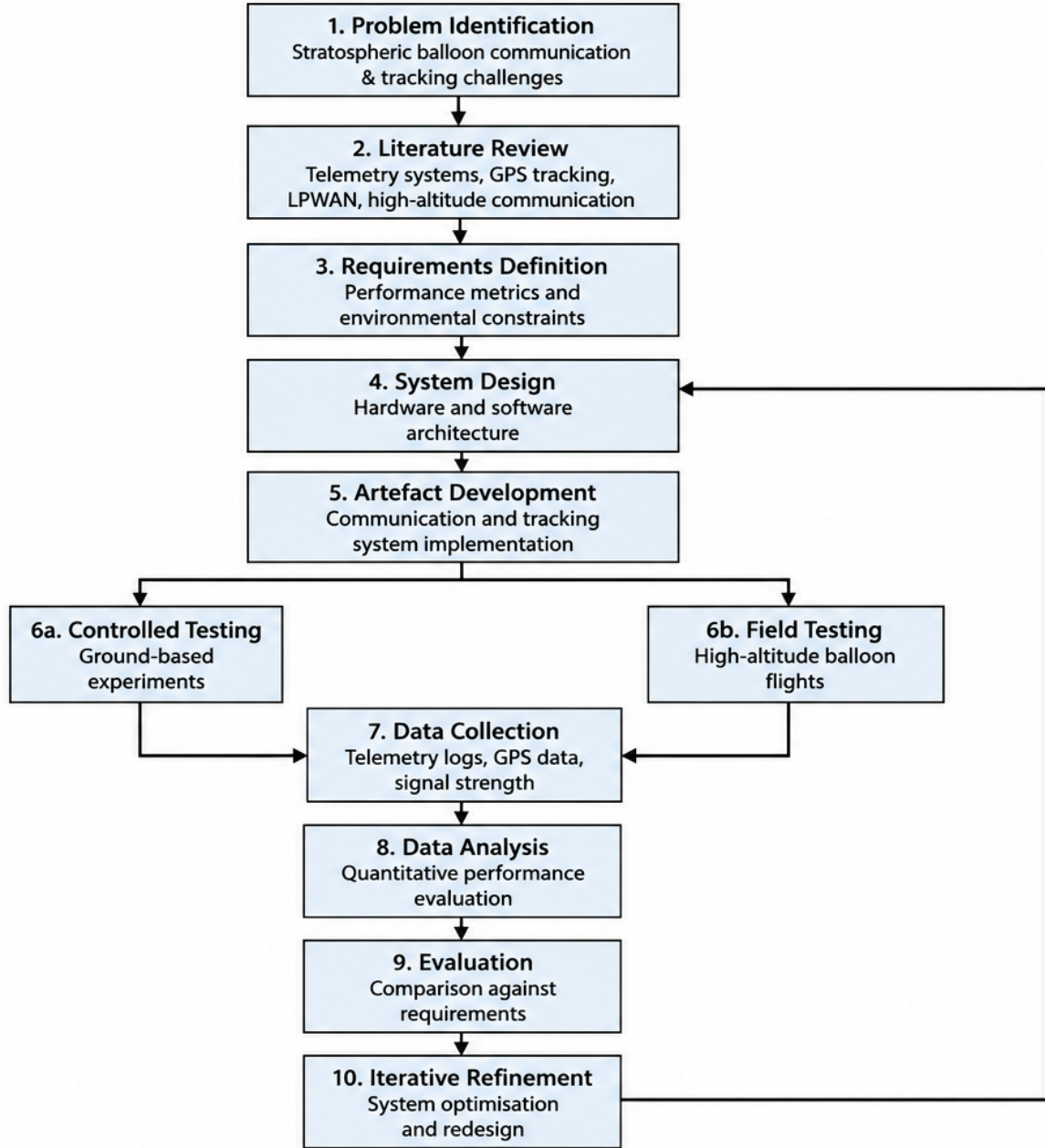


Figure 2.2: Research design followed in the study.

The artefact is evaluated through both controlled and real-world testing environments. Telemetry data collected during testing is analysed quantitatively to determine whether system objectives are achieved, with results used to refine the system iteratively [22, 23].

2.6 Conclusion

This chapter outlined the research methodology adopted for this study, including the pragmatic paradigm and DSR approach. It described the experimental and field testing methods used to evaluate system performance, as well as the quantitative techniques used for data analysis. The research design provides a structured framework for developing and evaluating the communication and tracking system under realistic operating conditions. The next chapter presents the literature review supporting the study.

Chapter 3

Literature review

3.1 Introduction

Reliable communication and tracking capabilities are fundamental requirements for high-altitude balloon systems because they enable telemetry transmission, position monitoring, efficient system operation, and payload recovery [25, 33, 35]. As a result, it is necessary to perform proper investigations on existing literature that will support and guide the development and completion of the proposed system.

This chapter presents a review of the literature used in this study. The reviewed literature provides insight into existing technologies, design approaches, operational limitations, and performance considerations associated with communication and tracking systems.

Literature relating to the key concepts central to this study is reviewed first, followed by additional literature relevant to the broader communication and tracking environment. Finally, the chapter discusses how the literature guides the design, implementation, and evaluation of the proposed communication and tracking system.

3.2 Related literature

The development of a communication and tracking system for a stratospheric balloon requires the integration of several technologies that must operate reliably within a demanding high-altitude environment [33, 35]. These technologies include telemetry systems for transmitting mission data, long-range communication systems for maintaining connectivity, and positioning systems for tracking and payload recovery [25, 35]. Existing research provides valuable insight into the capabilities, limitations, and design considerations associated with these technologies. Reviewing this literature establishes the theoretical foundation of the study, identifies proven approaches, and highlights challenges that must be addressed when developing a communication and tracking solution for stratospheric balloon applications.

This section reviews literature related to the key concepts identified in Chapter 1. The discussion focuses on technologies and design approaches relevant to the development of communication and tracking systems for high-altitude balloon applications.

3.2.1 Telemetry systems in high-altitude platforms

Telemetry systems are fundamental to the operation of high-altitude balloon missions because they provide the mechanism through which flight data, environmental measurements, and payload status information are communicated to ground operators [25, 35]. As balloon missions increasingly support scientific observation, environmental monitoring, and experimental communication systems, reliable telemetry has become essential for mission monitoring, operational decision-making, and payload recovery [25, 35]. Therefore, a considerable amount of research has focused on the design of telemetry systems capable of operating reliably over large distances while remaining energy efficient and lightweight [15, 25].

Telemetry systems generally consist of data acquisition, data processing, and communication subsystems that collectively enable the collection and transmission of information from remote platforms [25, 35]. In high-altitude balloon applications, telemetry data commonly includes GPS coordinates, altitude measurements, environmental sensor readings, battery status information, and system health indicators.

Puričar et al. describe telemetry systems specifically developed for stratospheric balloon applications and emphasise the importance of maintaining communication throughout all mission phases [25]. The authors highlight that telemetry reliability directly influences flight monitoring capabilities and the probability of successful payload recovery.

Wiyagi et al. present a high-altitude balloon payload architecture comprising sensing, processing, communication, positioning, and power management subsystems [35]. Their work illustrates the central role telemetry systems play in integrating data generated by multiple sensors and transmitting the resulting information to ground stations. The architecture further demonstrates the multidisciplinary nature of high-altitude payload design, where telemetry acts as the interface between onboard systems and mission operators.

Brown developed a multi-mode telemetry transmitter for high-altitude balloon applications and demonstrated the importance of communication redundancy in improving mission reliability [3]. The study showed that employing multiple communication mechanisms increases the probability of maintaining connectivity throughout the flight and improves payload recovery success rates.

Recent developments in telemetry communication have explored the use of low-power wide-area networking technologies. Krueger and Bergmaier investigated the use of LoRaWAN as a telemetry solution for high-altitude balloons and reported successful reception of more than 95% of transmitted telemetry data through a public LPWAN infrastructure [15]. The study demonstrated that modern telemetry systems can achieve reliable communication while simultaneously reducing system size, weight, and power consumption.

Experimental results presented by Krueger and Bergmaier further indicate that telemetry communication remained reliable at altitudes exceeding 2.6×10^1 km, with minimal packet loss recorded during flight testing [15]. These findings suggest that LPWAN technologies provide a viable alternative to traditional telemetry solutions for high-altitude applications.

The reviewed literature demonstrates that telemetry systems are critical components of high-altitude balloon missions. Reliable telemetry communication enables flight monitoring, supports mission operations, and improves payload recovery capabilities. Research further highlights the growing adoption of low-power communication technologies that provide long-range telemetry performance while maintaining low energy consumption, which also motivates the selection of LPWAN technology.

3.2.2 UHF-based telemetry communication systems

One of the primary challenges associated with high-altitude telemetry communication is maintaining reliable connectivity over large distances while operating within strict payload constraints. Communication technologies used in balloon systems must therefore provide long communication ranges, low power consumption, and acceptable reliability despite limited spectrum availability [16, 24]. These requirements motivated significant research into UHF and sub-GHz communication technologies, particularly LPWAN systems that are specifically designed for long-range, low-data-rate applications.

The extensive operational areas associated with high-altitude balloon missions require communication systems capable of maintaining connectivity across large geographical regions [20]. Traditional short-range wireless technologies often fail to meet these requirements due to limited transmission ranges and relatively high power consumption [10, 24]. Consequently, sub-GHz communication technologies have emerged as attractive alternatives for telemetry applications.

De Poorter et al. identified sub-GHz communication systems as an important class of LPWAN technologies capable of achieving communication ranges between 1 km and 50 km while maintaining low energy consumption [24]. According to the authors, the favourable propagation characteristics of lower frequencies enable larger coverage areas and improved penetration compared to higher-frequency alternatives.

The migration from the 2.4 GHz spectrum toward sub-GHz frequencies has largely been driven by the need for extended communication range and improved energy efficiency [24]. Although the study primarily focuses on 868 MHz and 900 MHz technologies, the authors indicate that the same communication principles apply to other sub-GHz frequencies, including 433 MHz [24].

A defining characteristic of LPWAN systems is the trade-off between communication range and throughput. De Poorter et al. explain that LPWAN technologies achieve long-range communication by operating at relatively low data rates while optimising power efficiency [24]. Technologies such as LoRaWAN further employ adaptive data rate (ADR) mechanisms to dynamically balance communication range and throughput according to prevailing operating conditions [24].

Bandwidth limitations remain a fundamental characteristic of LPWAN systems. According to De Poorter et al., these technologies are generally designed for small information exchanges involving payloads often smaller than 100 bytes [24]. Consequently, telemetry systems must carefully prioritise the information transmitted during flight operations.

Interference and coexistence challenges also influence LPWAN performance. De Poorter et al. note that limited spectrum availability and increasing adoption of LPWAN technologies can contribute to congestion and communication degradation within unlicensed sub-GHz frequency bands [24]. These considerations necessitate careful communication system design to maintain reliability under congested operating conditions.

The practical advantages of LPWAN communication are validated by Krueger and Bergmaier, who demonstrate that LoRaWAN provides significantly greater communication range than conventional wireless technologies while maintaining low power requirements [15]. Their flight experiments further confirm successful telemetry transmission from altitudes exceeding 2.6×10^1 km using sub-GHz communication technology [15].

The literature demonstrates that UHF and sub-GHz communication technologies provide an effective balance between communication range, energy efficiency, and system complexity. Although bandwidth limitations and interference remain important considerations, LPWAN technologies provide a suitable foundation for long-range telemetry communication in high-altitude balloon applications.

3.2.3 GPS-based tracking systems

Accurate position determination is essential during the flight of a high-altitude balloon. Position information supports flight monitoring, enables trajectory analysis, assists operational decision-making, and facilitates payload recovery following landing [6, 13]. As a result, GPS and other global navigation satellite system (GNSS) technologies have become central components of modern balloon tracking systems [6, 35] and require thorough research for successful tracking and monitoring of a stratospheric balloon system.

Enge describes GPS as a satellite-based positioning technology that determines receiver location through timing measurements obtained from multiple satellites [6]. GPS has become the dominant positioning technology because it offers global coverage, relatively high accuracy, and widespread availability [6].

Wing evaluated the performance of consumer-grade GPS receivers and identified several factors affecting positioning accuracy, including receiver quality, antenna design, satellite visibility, and environmental conditions [34]. These considerations remain relevant when selecting GPS hardware for high-altitude balloon applications.

Wiyagi et al. incorporated a Ublox NEO-M8N GNSS receiver into a high-altitude balloon payload and highlighted its suitability for operation at altitudes approaching 5×10^1 km [35]. The study demonstrates the practical feasibility of integrating GNSS technologies into lightweight atmospheric research platforms.

Recent developments have increasingly integrated GNSS positioning with LPWAN communication technologies. Janssen et al. describe architectures in which GNSS observations are transmitted through LPWAN networks to cloud-based processing platforms for position determination [13]. Such approaches reduce computational requirements on resource-constrained devices while improving energy efficiency.

Janssen et al. further discuss integrated GNSS and LPWAN systems-on-chip that combine positioning and communication capabilities within a single device [13]. These developments reflect a growing convergence between communication and positioning technologies in modern IoT systems.

The survey also highlights positioning architectures in which communication networks serve as transport mechanisms for GNSS observations and assistance data [13]. Consequently, positioning performance increasingly depends not only on GPS receiver capabilities but also on the reliability of supporting communication systems.

The reviewed literature confirms that GPS remains the most suitable positioning technology for high-altitude balloon applications. However, effective tracking increasingly depends on the integration of position tracking and communication technologies, highlighting the importance of reliable telemetry systems for successful tracking performance.

3.2.4 Additional considerations for high-altitude tracking and communication systems

While telemetry communication and GPS-based tracking form the core components of the proposed system, several additional factors influence the overall effectiveness, reliability, and operational success of high-altitude balloon missions. Previous research has identified challenges related to payload recovery, communication coverage, system integration, and technology selection, all of which affect the design of communication and tracking solutions [3, 4, 35]. The following literature therefore examines complementary considerations that provide further context for the development of a robust and practical stratospheric balloon subsystem.

Hybrid tracking and recovery systems

The recovery of high-altitude balloon payloads remains a critical operational objective because payloads often contain valuable scientific instruments, experimental hardware, and recorded mission data [3, 8]. Consequently, researchers have increasingly investigated hybrid tracking approaches that improve reliability through redundancy.

Brown highlights the importance of communication redundancy through the implementation of multi-mode telemetry systems [3]. By employing multiple communication channels, such systems reduce the likelihood of total communication failure and improve payload recovery success rates.

Janssen et al. discuss hybrid architectures that combine GNSS positioning with LPWAN, cellular, and low earth orbit (LEO) satellite communication technologies [13]. These approaches improve coverage and tracking reliability, particularly in remote environments where individual communication technologies may experience limitations.

The authors further describe multi-radio access technology architectures that dynamically select communication mechanisms according to environmental conditions, energy constraints, and coverage availability [13]. Such adaptive systems improve operational resilience while maintaining efficient resource utilisation.

The reviewed studies demonstrate that existing recovery-oriented systems frequently rely on multiple communication technologies to improve tracking reliability. These approaches provide useful benchmarks for evaluating the proposed system, particularly with respect to communication redundancy, tracking continuity, and payload recovery effectiveness.

Limitations of cellular communication for high-altitude platforms

Although cellular communication networks provide extensive terrestrial coverage, their suitability for high-altitude applications remains limited. Cellular infrastructure is primarily designed to support ground-based users and therefore does not adequately account for aerial operating environments.

Colpaert et al. demonstrate that aerial platforms experience significantly different radio propagation conditions compared to terrestrial users [4]. At increasing altitudes, aerial platforms become exposed to signals from multiple base stations simultaneously, resulting in substantial inter-cell interference.

Simulation results presented by Colpaert et al. indicate that signal quality decreases significantly with altitude because increasing numbers of base stations maintain line-of-sight communication paths to the aerial platform [4]. This creates interference conditions that degrade communication reliability.

Coverage analyses further demonstrate that communication performance deteriorates rapidly with increasing altitude [4]. The authors conclude that conventional LTE and mmWave systems cannot reliably support aerial users at elevated operating heights without substantial modifications.

These findings are particularly relevant for stratospheric balloon systems because operational altitudes significantly exceed those considered during conventional cellular network planning [4].

Integrated communication and tracking architectures

Communication and tracking systems are inherently multidisciplinary because they require the integration of sensing, processing, positioning, power management, and communication subsystems [35]. Consequently, system architecture plays an important role in determining overall system performance and reliability [35].

Wiyagi et al. present a comprehensive balloon payload architecture comprising of sensors, microcontrollers, a Raspberry Pi processing platform, communication modules, and GPS positioning systems [35]. Figure 3.1 illustrates how multiple subsystems can be integrated to support mission objectives while operating within strict payload constraints.

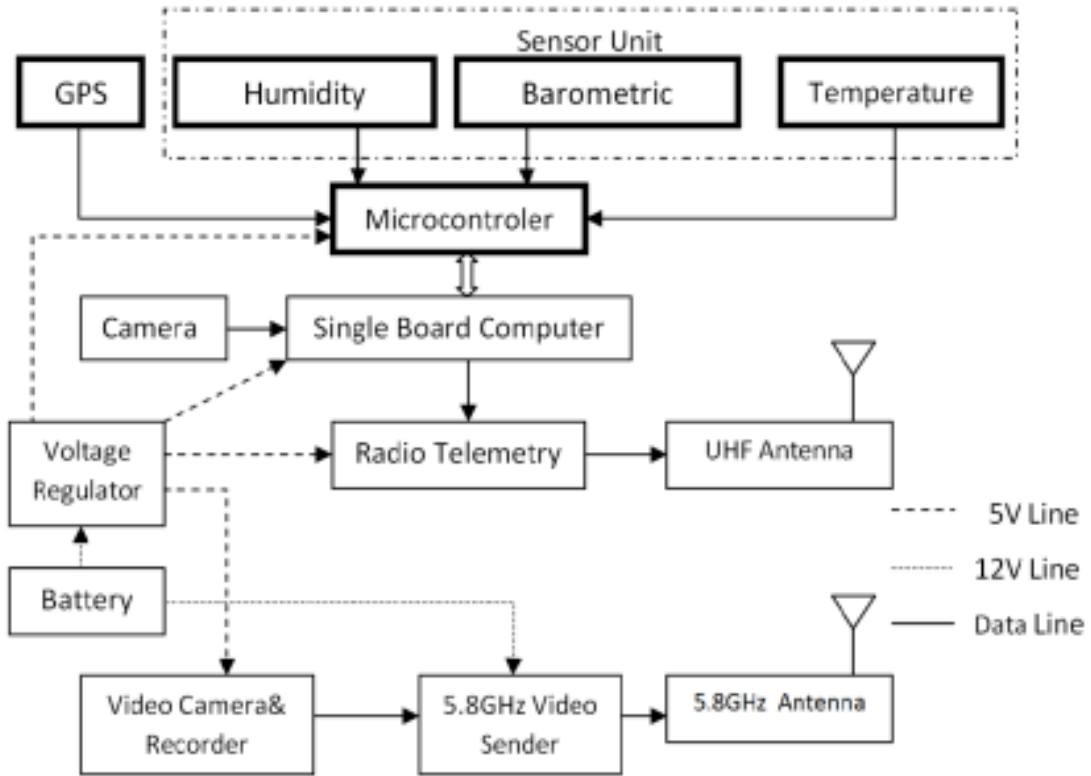


Figure 3.1: High-altitude balloon payload architecture proposed by Wiyagi et al. [35].

The use of Raspberry Pi platforms for data processing and communication management further demonstrates the suitability of modular architectures for high-altitude systems [35]. Modular architectures facilitate subsystem integration, improve maintainability, and support future system expansion.

Comparable architectural principles are evident in IoT-based monitoring systems. Uzair et al. describe an integrated architecture in which Arduino-based sensing platforms communicate with Raspberry Pi processing systems through layered communication structures [32]. The separation of sensing, processing, communication, and application functions provides a useful reference for communication and tracking system design.

Several studies have implemented integrated high-altitude balloon payloads that combine communication, positioning, processing, and sensing functions within a single system. Examples include the architectures proposed by Wiyagi et al. [35] and Wang et al. [33], both of which demonstrate the practical integration of telemetry and tracking subsystems in high-altitude environments. These studies provide reference systems against which the proposed communication and tracking solution can be compared.

3.3 How the literature guides this study

The literature reviewed in this chapter provides the theoretical and technical foundation upon which the proposed communication and tracking system is developed. Beyond establishing the current state of research, the reviewed studies identify key design requirements, technological constraints, and performance considerations that directly influence the development of the proposed subsystem.

The reviewed research supports the selection of a sub-GHz communication approach for telemetry transmission. Studies demonstrate that LPWAN technologies offer a practical balance between communication range, power consumption, and system complexity, making them suitable for high-altitude balloon applications [15, 24]. At the same time, the identified limitations relating to bandwidth availability, interference, and spectrum utilisation highlight important constraints that must be considered during system design and implementation.

The literature also informs the selection of GPS as the primary tracking mechanism. Previous studies demonstrate the suitability of GNSS technologies for high-altitude applications and emphasise the growing integration between communication and positioning systems [6, 13]. These findings support the development of a tracking solution in which positional information forms an integral component of the telemetry data transmitted to ground operators.

Research relating to system architecture further guides the organisation of the proposed subsystem [32, 33, 35]. Existing payload designs demonstrate the value of integrating sensing, processing, positioning, communication, and power-management functions into a modular architecture [32, 33, 35]. These architectural principles influence the selection and interaction of the hardware and software components implemented in this study.

In addition to informing design decisions, the reviewed literature establishes a basis for evaluating the developed artefact. Performance measures reported in previous studies, including communication reliability, packet delivery performance, operational range, and tracking effectiveness, provide reference points against which the proposed system can be assessed [3, 13, 15]. The findings presented in subsequent chapters can therefore be compared with those reported in existing research to determine whether the developed solution achieves its intended objectives.

The literature review therefore serves as both a knowledge base and a design guide for the study. It provides insight into the selection of technologies, identifies relevant design considerations, and establishes evaluation criteria that support the systematic development and assessment of the proposed communication and tracking system.

3.4 Conclusion

This chapter reviewed literature relevant to telemetry systems, UHF-based communication technologies, GPS-based tracking systems, and additional considerations affecting communication and tracking performance in high-altitude balloon applications. The reviewed literature established the theoretical and technological foundation necessary for the development of a communication and tracking system capable of operating under stratospheric conditions.

The literature demonstrates that successful high-altitude balloon missions require reliable long-range communication, accurate positioning, efficient energy utilisation, and robust payload recovery mechanisms [3, 25, 35]. Existing studies support the use of sub-GHz communication technologies and GPS-based tracking systems while highlighting the limitations of cellular communication at high altitudes [4, 6, 15, 24]. The reviewed literature further emphasises the value of integrated system architectures and communication redundancy in improving overall mission reliability.

The next chapter presents the empirical design and development process used to design and develop the proposed communication and tracking system. It outlines the system artefact, describes the iterative development cycles, and explains the evaluation procedures used to assess performance and reliability.

Chapter 4

Empirical study

4.1 Introduction

In the mini-dissertation, the development of a reliable and power-efficient communication and tracking system for a stratospheric balloon operating at high altitudes is investigated. The context and objectives of this study were introduced in Chapter 1, Chapter 2 established the pragmatic paradigm, Design Science Research (DSR) methodology, and Chapter 3 reviewed the literature relevant to telemetry systems, UHF-based communication, and GPS-based tracking. This chapter addresses the design-and-development, empirical objective of the study: to design, implement, and evaluate the proposed communication and tracking artefact through the DSR process.

The empirical study chapter discusses the empirical design and development of the proposed communication and tracking system. It describes the artefact that was produced, explains how the DSR process phases (problem identification, objective definition, design and development, demonstration, and evaluation) were applied and iterated in practice, and reports the findings obtained from testing the proposed system.

The remainder of this chapter is structured as follows. The artefact description describes the communication and tracking artefact, including its purpose, scope, and key requirements, for both the airborne subsystem and the ground station. The description of the development process describes the development process followed, structured according to the DSR phases, and discusses the iterative cycles undertaken. The findings section presents the findings obtained from GPS-based testing and communication-link testing. Finally, this chapter concludes the empirical study of this mini-dissertation and prepares the reader for the conclusion chapter.

4.2 Artefact description

The artefact produced in this study is a communication and tracking system for a stratospheric balloon, developed as one of three interdependent subsystems within a broader group project. The purpose of the artefact is to provide reliable, low-power position tracking and telemetry transmission between a stratospheric balloon and a ground station, operating within the power, bandwidth, and environmental constraints of stratospheric flight [10, 11, 25]. Importantly, the artefact is not limited to the airborne payload; it also includes the design, implementation, and testing of the ground station that receives, logs, and displays the transmitted telemetry, since reliable end-to-end communication depends equally on both ends of the link.

4.2.1 Purpose and scope

The development of the artefact is motivated by the problem identified in Chapter 1: conventional terrestrial infrastructure, such as cellular networks, is unsuitable for high-altitude communication because it is designed for ground-based coverage and suffers from severe inter-cell interference and signal degradation at altitude [4]. UHF communication in the 433 MHz ISM band, by contrast, offers long-range, low-power communication that tolerates the line-of-sight and propagation conditions typical of stratospheric flight [16, 24]. The artefact therefore combines a UHF telemetry link with GPS-based positioning to meet the aims defined in Section 1.3.3, and pairs the airborne payload with a dedicated ground station capable of reliably receiving and recording this data throughout the flight.

The scope of the artefact covers both segments of the communication and tracking link. On the airborne side, it is responsible for acquiring GPS position data, receiving processed data from the control and sensor subsystems developed by the other two collaborators in the group project, packaging this data into a telemetry format, and transmitting it over a UHF link. On the ground side, the scope includes the design, assembly, and testing of a ground station capable of receiving the UHF telemetry signal, decoding the transmitted packets, and logging the resulting position data for later analysis against the evaluation metrics defined in Section 1.4.4. Onboard control logic, payload release, geofencing, and atmospheric sensing fall outside of the scope for this artefact and are addressed by the other subsystems (Section 1.4.6).

4.2.2 System architecture

The artefact consists of an airborne subsystem and a ground station subsystem. The following components were selected for their low power consumption, reliability, and suitability for long-range telemetry, consistent with prior stratospheric balloon and LPWAN research [24, 32, 35]:

- A **GPS module** (Uputronics) for acquiring real-time positional data (latitude, longitude, altitude) [6].
- A **Raspberry Pi Zero** single-board computer, responsible for reading GPS data, formatting telemetry packets, and interfacing with the UHF transmitter.

- A **Radiometrix NTX2-434.650-10-HP UHF transmitter**, mounted on the balloon payload and operating at 434.650 MHz within the 433 MHz ISM band, providing the radio-frequency communication link to the ground station [26].
- A **Radiometrix NRX2 UHF receiver**, mounted at the ground station, together with a computer for decoding, displaying, and logging received telemetry packets [26].
- A battery-based power supply for the airborne subsystem.

Together, these components form the two ends of the communication and tracking link: the airborne payload, which acquires and transmits telemetry, and the ground station, which receives, decodes, and logs it for monitoring and recovery purposes [25].

4.2.3 System requirements

Based on the aims and objectives defined in Section 1.3.3, the following key requirements guided the design of the artefact:

- The system must be able to acquire and transmit GPS location data to a ground station at regular intervals,
- Operate within the power constraints of a battery-powered, high-altitude platform,
- Provide a communication range sufficient for stratospheric operating altitudes,
- Be capable of reliably receiving, decoding, and logging telemetry data transmitted from the balloon throughout the flight,
- Support redundancy mechanisms to improve the likelihood of payload recovery in the event of communication failure (Section 1.5.3) [3] and,
- Minimise transmitted information to reduce the risk of unintended information leakage [17], as discussed in Section 1.5.1.

These requirements collectively guided the design and testing decisions described in the remainder of this chapter, for both the airborne payload and the ground station.

4.3 Description of the development process

This section describes how the DSR process phases were applied in practice during the development of the communication and tracking artefact, covering both the airborne payload and the ground station. Development was iterative: each cycle below concludes with a reflection describing what was learned and how it informed the next cycle, consistent with the rigour and validity procedures described in Section 2.3.3.

4.3.1 Problem identification and objective definition

The communication and tracking subsystem was developed to address the need for reliable position tracking and telemetry communication for the stratospheric balloon. The system must be capable of obtaining the balloon's geographic position and making this information available for transmission to the ground station while operating within the constraints of a high-altitude payload. The development therefore focused on establishing a functional interface between the GPS module and the onboard Raspberry Pi Zero, integrating the acquired positional information into the telemetry system, and ultimately transmitting the resulting data to the ground station through the UHF communication link.

The development objectives were translated into a series of practical implementation requirements. The first requirement was to establish reliable communication between the GPS module and Raspberry Pi Zero and obtain valid positional information. Once this interface was established, the GPS data needed to be made available to other components of the system, including the geofencing functionality and the telemetry transmission process. The UHF transmitter and ground station then needed to be integrated to provide an end-to-end communication link between the airborne and ground-based components.

4.3.2 Design and development cycle 1

The first development cycle focused on establishing communication between the Uputronics GPS module and the Raspberry Pi Zero. This was selected as the initial development activity because GPS position information forms the basis of the tracking functionality and is required for both the geofencing functionality and telemetry transmission. The objective of this cycle was therefore to establish a functional interface through which the Raspberry Pi Zero could obtain positional information from the GPS module.

A challenge during the initial implementation was determining the required wiring and interface configuration for the GPS module. Documentation for the specific GPS module was not readily available, making it necessary to investigate the available connections and determine which pins were required for communication with the Raspberry Pi Zero. Via trial and error and moderate experience with Raspberry Pi devices, the required connections were identified and configured to establish communication between the GPS module and the Raspberry Pi Zero as shown in Figure-4.1.

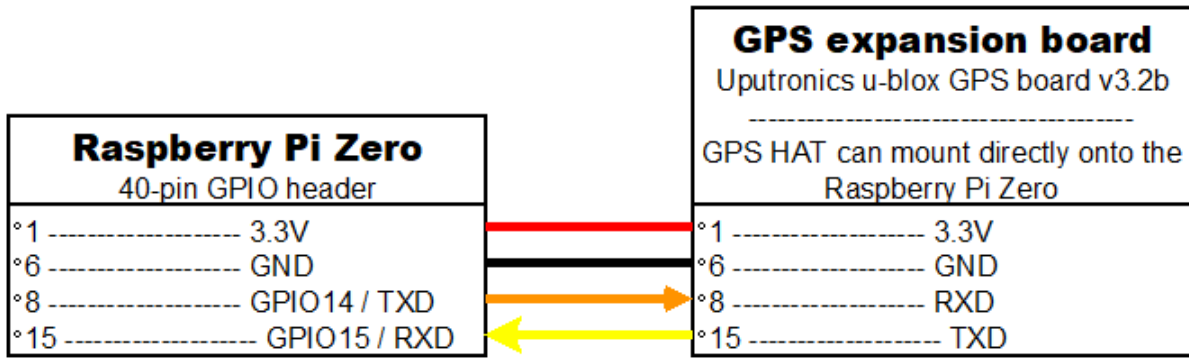


Figure 4.1: Verified UART wiring configuration between the Raspberry Pi Zero and Uputronics GPS expansion board.

After the wiring configuration was determined, a program was developed to communicate with the GPS module and retrieve its positional information. The program implemented for the GPS module to be able to interface with the Raspberry Pi Zero is provided in Appendix A.1; Listing 4.1 below shows the validation function used to determine whether a reported GPS fix is complete enough to be logged or used by the rest of the system.

```

35 def is_complete(report):
36     mode = getattr(report, "mode", 0)
37
38     if (mode < MIN_MODE):
39         return False, f"mode={mode} (need >= {MIN_MODE})"
40
41     missing = [
42         field for field in REQUIRED_FIELDS
43         if getattr(report, field, None) is None
44     ]
45
46     if (missing):
47         return False, f"missing: {' '.join(missing)}"
48     return True, None

```

Listing 4.1: Validity check applied to each GPS fix before it is logged or made available to other components.

The program executed successfully, processed the GPS data and displayed the results in the Raspberry Pi Zero terminal. As shown in Figure 4.2, the program continuously reported valid GPS fixes, including latitude, longitude, altitude, speed, track, and the number of satellites used. This demonstrates that the Raspberry Pi Zero can successfully communicate with the GPS module and acquire positional data for further processing.

```

strato@StratoBalloon:~/GPS/GroundTest $ sudo python3 gps_execute.py
GPS monitoring started.
Logging to: /home/strato/GPS/GroundTest/logs/gps_log_2026-09-08_22-05-02.csv
Latest GPS data: /home/strato/GPS/GroundTest/latest_gps.json
Valid GPS fixes are logged every 5 seconds.
Press Ctrl+C to stop.

SYSTEM TIME: 2026-09-08T22:05:03.158387+00:00 | SATELLITE TIME: 2026-09-08T22:05:03.000Z
VALID FIX LOGGED lat=-26.688736 lon=27.095180 alt=1322.971m speed=0.061m/s track=166.6631deg sats_used=0
SYSTEM TIME: 2026-09-08T22:05:09.159207+00:00 | SATELLITE TIME: 2026-09-08T22:05:09.000Z
VALID FIX LOGGED lat=-26.688705 lon=27.095179 alt=1322.144m speed=0.149m/s track=166.6631deg sats_used=6
SYSTEM TIME: 2026-09-08T22:05:14.157966+00:00 | SATELLITE TIME: 2026-09-08T22:05:14.000Z
VALID FIX LOGGED lat=-26.688699 lon=27.095177 alt=1319.697m speed=0.028m/s track=166.6631deg sats_used=6
SYSTEM TIME: 2026-09-08T22:05:19.623612+00:00 | SATELLITE TIME: 2026-09-08T22:05:19.000Z
VALID FIX LOGGED lat=-26.688678 lon=27.095158 alt=1316.164m speed=0.217m/s track=166.6631deg sats_used=7
SYSTEM TIME: 2026-09-08T22:05:25.159361+00:00 | SATELLITE TIME: 2026-09-08T22:05:25.000Z
VALID FIX LOGGED lat=-26.688675 lon=27.095138 alt=1310.473m speed=0.207m/s track=166.6631deg sats_used=6
SYSTEM TIME: 2026-09-08T22:05:30.165357+00:00 | SATELLITE TIME: 2026-09-08T22:05:30.000Z
VALID FIX LOGGED lat=-26.688646 lon=27.095130 alt=1310.008m speed=0.399m/s track=166.6631deg sats_used=5
SYSTEM TIME: 2026-09-08T22:05:35.622294+00:00 | SATELLITE TIME: 2026-09-08T22:05:35.000Z
VALID FIX LOGGED lat=-26.688657 lon=27.095135 alt=1309.486m speed=0.306m/s track=166.6631deg sats_used=6
SYSTEM TIME: 2026-09-08T22:05:41.160143+00:00 | SATELLITE TIME: 2026-09-08T22:05:41.000Z
VALID FIX LOGGED lat=-26.688679 lon=27.095149 alt=1310.31m speed=0.078m/s track=166.6631deg sats_used=7
SYSTEM TIME: 2026-09-08T22:05:47.157988+00:00 | SATELLITE TIME: 2026-09-08T22:05:47.000Z
VALID FIX LOGGED lat=-26.688759 lon=27.095155 alt=1307.48m speed=0.579m/s track=166.6631deg sats_used=5
SYSTEM TIME: 2026-09-08T22:05:52.161992+00:00 | SATELLITE TIME: 2026-09-08T22:05:52.000Z
VALID FIX LOGGED lat=-26.688832 lon=27.095170 alt=1306.375m speed=0.876m/s track=166.7016deg sats_used=7
SYSTEM TIME: 2026-09-08T22:05:57.620368+00:00 | SATELLITE TIME: 2026-09-08T22:05:57.000Z
VALID FIX LOGGED lat=-26.688826 lon=27.095161 alt=1305.13m speed=0.481m/s track=171.4333deg sats_used=6
SYSTEM TIME: 2026-09-08T22:06:03.159622+00:00 | SATELLITE TIME: 2026-09-08T22:06:03.000Z
VALID FIX LOGGED lat=-26.688854 lon=27.095163 alt=1303.98m speed=0.507m/s track=171.4333deg sats_used=6
SYSTEM TIME: 2026-09-08T22:06:09.162545+00:00 | SATELLITE TIME: 2026-09-08T22:06:09.000Z
VALID FIX LOGGED lat=-26.688844 lon=27.095164 alt=1306.038m speed=0.347m/s track=171.4333deg sats_used=7
SYSTEM TIME: 2026-09-08T22:06:15.158881+00:00 | SATELLITE TIME: 2026-09-08T22:06:15.000Z
VALID FIX LOGGED lat=-26.688849 lon=27.095185 alt=1310.426m speed=0.396m/s track=171.4333deg sats_used=8
SYSTEM TIME: 2026-09-08T22:06:20.159128+00:00 | SATELLITE TIME: 2026-09-08T22:06:20.000Z
VALID FIX LOGGED lat=-26.688839 lon=27.095197 alt=1313.702m speed=0.319m/s track=171.4333deg sats_used=8
SYSTEM TIME: 2026-09-08T22:06:25.621640+00:00 | SATELLITE TIME: 2026-09-08T22:06:25.000Z
VALID FIX LOGGED lat=-26.688851 lon=27.095214 alt=1316.588m speed=0.439m/s track=171.4333deg sats_used=6

```

Figure 4.2: Example of successful GPS data acquisition, showing consecutive valid GPS fixes containing latitude, longitude, altitude, speed, track, and the number of satellites used.

In addition to recording the GPS observations, the latest valid GPS fix is made available to the entire system by storing the latest GPS data in a JSON file within the same directory as the Python program that interfaces with the GPS module. This allows other components to access the most recent positional information without requiring a separate direct connection to the GPS module. Thus, the developed Python program establishes the positional data source required for subsequent integration with the geofencing and telemetry transmission components. Additionally, the Python GPS program logs each positional fix at regular intervals to allow for quantitative analysis in Section 4.4.1.

Reflection on this cycle indicates that establishing and verifying the GPS interface before integrating additional components reduces the complexity of further development. The verified GPS connection provides a known working baseline for the next development cycle, where the payload transmitter will transmit GPS data to the ground station receiver. The wiring configuration and successful GPS output were also documented to provide a reproducible configuration for development and testing.

4.3.3 Design and development cycle 2

Building on the verified GPS interface in Cycle 1, this cycle focused on integrating the UHF transmission component of the communication link. Starting with the physical wiring of the Radiometrix NTX2 UHF transmitter to the Raspberry Pi Zero. Establishing the transmitter interface independently of the GPS subsystem allows for the transmission subsystem to be configured and verified before integrating it with the GPS data source.

The NTX2 transmitter is wired following a UKHAS community wiring diagram for interfacing the NTX2 with a microcontroller [30], together with the manufacturer’s datasheet [26], as shown in Figure 4.3.

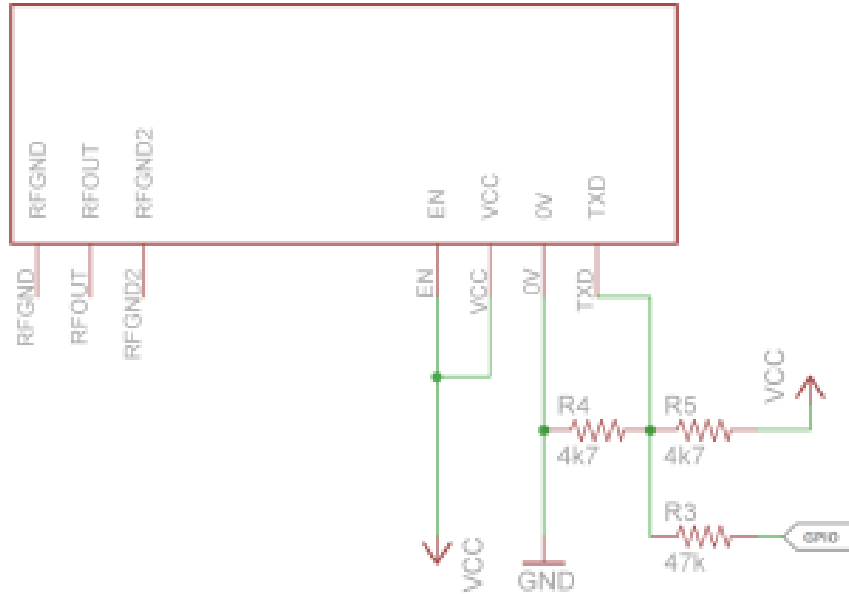


Figure 4.3: Wiring configuration between the Raspberry Pi Zero and the Radiometrix NTX2 UHF transmitter, based on [30].

The series resistor, R3, used on the TXD input was evaluated as part of the development of the NTX2 transmitter interface. The initial resistor configuration was based on the UKHAS guide for interfacing the NTX2 with a microcontroller, which specifies a 47 k Ω resistor in its example circuit [30]. This value was therefore used as the initial implementation value based on the community implementation guide and was not treated as a value specified by the Radiometrix datasheet. The manufacturer’s documentation for the NTX2/NRX2 was consulted separately to verify the electrical characteristics and operating requirements of the actual transmitter and receiver used in the artefact [26].

During development, the R3 value was adjusted experimentally to determine a suitable interface configuration between the Raspberry Pi Zero and the NTX2 transmitter. This was necessary because the resistor value affects the electrical signal applied to the transmitter's TXD input as well as the modulation of the transmitted signal. The 1 k Ω resistor was selected for the final implementation based on the observed performance of the complete transmitter and receiver system. Alternative resistor values were tested during development, including a 470 Ω configuration, which produced poorer communication performance than the 1 k Ω configuration. The 1 k Ω value was therefore kept as an empirically selected design value for R3, rather than being presented as a value directly calculated from or specified by the manufacturer. This iterative adjustment formed part of the development process used to establish a configuration that provided suitable signal behaviour for subsequent testing of the NTX2-to-NRX2 communication link. This produced the completed transmitter artefact, as shown in Figure 4.4.

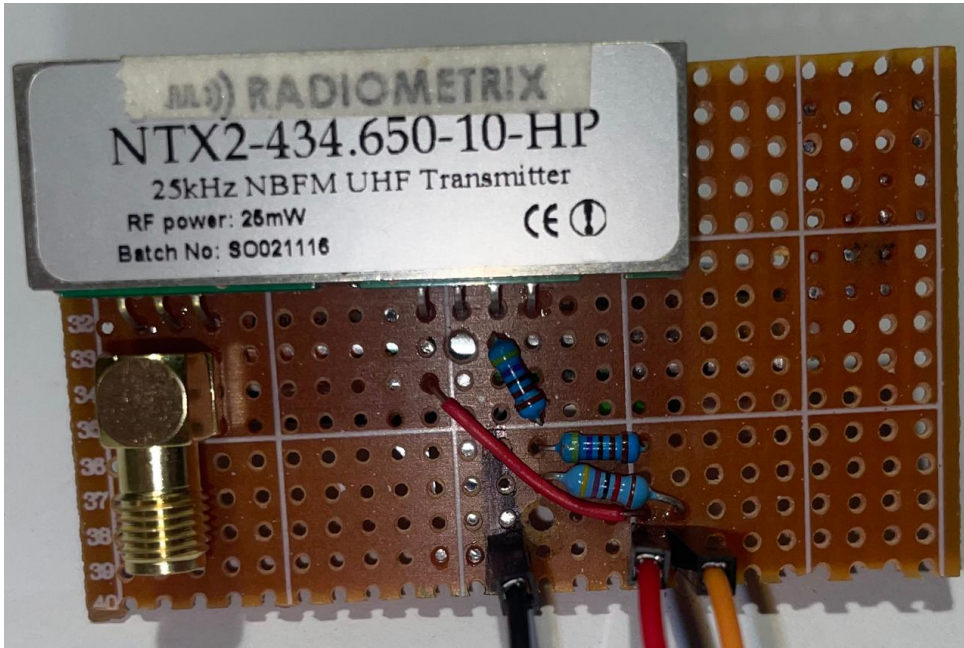


Figure 4.4: Completed Radiometrix NTX2 UHF transmitter artefact.

After establishing the wiring for the NTX2 transmitter, the 434.650,MHz antenna required for transmission was also constructed. A quarter-wave ground-plane antenna was built, consisting of a central radiating element connected through a 50 Ω coaxial cable [26], together with four ground-plane radial wires arranged radially and cut to 16.4,cm according to the practical antenna dimensions specified by the UKHAS community guide [18, 31].

[Requires developed antennas and testing results/programs]

4.3.4 Ground station design and testing

An Arduino Uno R3 was selected as the microcontroller used to interface with the Radiometrix NRX2 receiver because its available digital I/O, serial communication capability, USB connectivity, and suitability for rapid prototyping were sufficient for the requirements of the ground station [1]. Like the NTX2 transmitter, the receiver produces a raw baseband output that requires a microcontroller to decode the received signal into usable telemetry data [26, 30]. The

required connections between the Radiometrix receiver and the Arduino Uno R3 were identified, as shown in Figure 4.5.

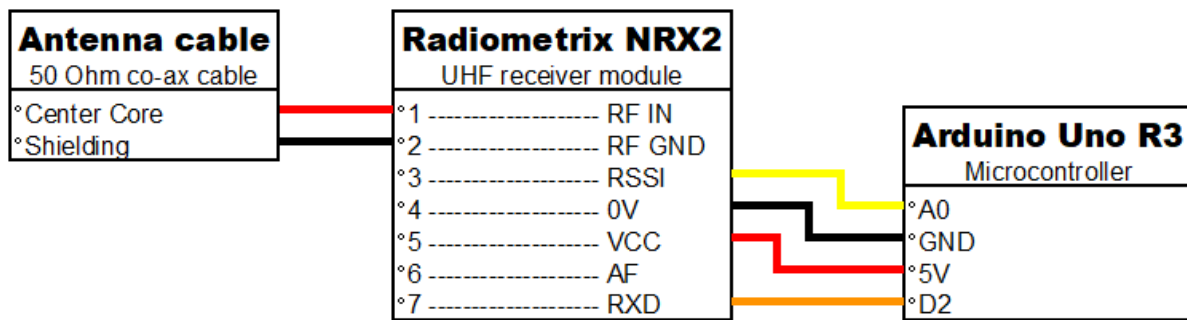


Figure 4.5: Wiring configuration between the Radiometrix NRX2 receiver and the Arduino Uno R3 at the ground station.

The antenna used at the ground station was constructed using the same quarter-wave ground-plane design described under Design and development cycle 2 above, consisting of a central radiating element connected via $50\ \Omega$ coaxial cable and four 16.4 cm ground-plane radial wires [26, 31].

[Requires developed antennas and testing results/programs]

4.4 Findings

This section evaluates the performance of the developed communication and tracking artefact against the objectives and evaluation metrics defined for the study. The results are presented for the GPS tracking component and, once completed, the end-to-end communication link between the airborne payload and ground station. GPS-based testing established the GPS receiver’s positional consistency under stationary conditions, while the subsequent field test will evaluate the GPS tracking component’s behaviour under flight conditions. The findings are used to determine whether the developed artefact provides the required functionality and performance for its intended stratospheric balloon application.

4.4.1 GPS-based testing results

GPS-based testing was conducted to evaluate the performance and consistency of the GPS component. The tests were performed at stationary locations to determine the extent to which the GPS module maintained a consistent reported position while connected to the Raspberry Pi Zero. The recorded GPS observations were analysed using the developed program included in Appendix A, which calculates positional deviations and changes between consecutive GPS observations from the recorded data.

The results obtained from the analysis are presented in Table 4.1 and calculated using the Python program in Appendix A.2. The table summarises the positional variation observed across 22 GPS-based test sessions. Mean and maximum positional deviation provide an indication of the overall variation in the reported position, while the mean consecutive position changes indicate the magnitude of movement between successive GPS observations. The number of position changes greater than 5m is included to identify larger deviations that occurred while the GPS receiver remained stationary.

Table 4.1: Quantitative summary of GPS-based GPS positional consistency.

Metric	Result
Number of test sessions	22
Total GPS readings	13356
Mean positional deviation (m)	5.623
Maximum positional deviation (m)	51.475
Mean consecutive position change (m)	1.618
Position changes >5 m	93

The values shown in Table 4.1 were obtained from the terminal output produced when the analysis program was executed. The program processed the recorded GPS observations from the 22 GPS-based test sessions and calculated the positional consistency metrics presented in the table. The terminal output provided a direct record of the calculated results and was used to verify that the values reported in Table 4.1 corresponded to the processed test data. Figure 4.6 shows an example of the terminal output generated during the analysis.

Log file	Readings	Mean Dev.	Max Dev.	Mean Change	Max Change	Jumps
gps_log_2026-09-08_22-10-59.csv	61	7.651 m	16.017 m	1.512 m	5.204 m	1
gps_log_2026-09-08_22-16-32.csv	59	6.071 m	25.039 m	1.937 m	11.331 m	5
gps_log_2026-09-08_22-21-59.csv	55	5.761 m	15.069 m	1.691 m	9.289 m	2
gps_log_2026-09-08_22-39-08.csv	65	7.859 m	23.341 m	2.162 m	10.446 m	6
gps_log_2026-09-09_00-44-00.csv	411	4.217 m	21.898 m	0.759 m	6.678 m	2
gps_log_2026-09-09_01-24-39.csv	5	0.112 m	0.233 m	0.126 m	0.244 m	0
gps_log_2026-09-09_01-25-23.csv	4036	4.856 m	15.985 m	0.468 m	2.985 m	0
gps_log_2026-09-13_16-42-21.csv	2949	8.688 m	51.475 m	1.385 m	15.692 m	55
gps_log_2026-09-14_07-01-11.csv	2995	4.857 m	27.727 m	0.795 m	7.090 m	5
gps_log_2026-09-14_12-44-34.csv	1046	4.272 m	13.934 m	0.603 m	2.651 m	0
gps_log_2026-09-14_16-48-37.csv	732	6.488 m	22.487 m	1.104 m	7.357 m	4
gps_log_2026-09-14_17-58-01.csv	572	4.587 m	13.139 m	0.856 m	4.173 m	0
gps_log_2026-09-16_20-57-11.csv	23	7.562 m	16.097 m	3.586 m	15.071 m	5
gps_log_2026-09-16_21-02-34.csv	16	15.061 m	36.847 m	4.807 m	11.139 m	5
gps_log_2026-09-16_21-04-07.csv	12	7.879 m	27.418 m	6.707 m	30.663 m	3
gps_log_2026-09-16_21-23-49.csv	6	0.641 m	1.164 m	0.535 m	0.855 m	0
gps_log_2026-09-16_21-26-07.csv	77	4.583 m	11.808 m	1.007 m	3.202 m	0
gps_log_2026-09-16_21-45-50.csv	17	4.455 m	8.985 m	1.504 m	3.256 m	0
gps_log_2026-09-16_22-46-27.csv	90	3.890 m	9.023 m	1.262 m	4.240 m	0
gps_log_2026-09-17_22-23-59.csv	60	8.034 m	14.000 m	0.900 m	3.089 m	0
gps_log_2026-09-17_22-34-54.csv	15	2.435 m	3.191 m	0.774 m	2.220 m	0
gps_log_2026-09-19_02-18-10.csv	54	3.741 m	9.940 m	1.109 m	2.744 m	0

OVERALL SUMMARY

Measurement	Value
Total CSV files read	22
Files containing GPS values	22
Empty CSV files	0
Total GPS readings	13356

GPS Metric	Result
Average positional deviation	5.623 m
Largest positional deviation	51.475 m
Average consecutive position change	1.618 m
Total jumps > 5 m	93

Figure 4.6: Terminal output produced by the GPS log analysis program.

The calculated results indicate that the GPS receiver exhibited measurable positional variation while remaining stationary. Across the three test sessions, the mean positional deviation was 5.623 m, with a maximum deviation of 51.475 m. The mean change between consecutive observations was 1.618 m. A total of 93 consecutive position changes exceeded 5m. These variations represent changes in the reported GPS position rather than physical movement of the receiver, and therefore provide an indication of the positional consistency that can be expected from the GPS component under stationary conditions [34].

The metrics in Table 4.1 were generated by the core analysis function shown in Listing 4.2. The function processes successive GPS observations, calculates the distance between reported positions, determines the positional deviations, and aggregates the resulting values into the summary statistics reported above.

```

34 def analyse_file(path):
35     #Analyse one GPS CSV file.
36     with open(path, newline="", encoding="utf-8") as f:
37         f.readline() # Skip metadata header line
38         reader = csv.DictReader(f)
39         if (not {"latitude", "longitude", "altitude_m", "speed_mps", "
40             satellites_used"}.issubset(reader.fieldnames or [])):
41             return None
42         rows = list(reader)
43
44     valid_rows = []
45     for row in rows:
46         try:
47             valid_rows.append((row, float(row["latitude"]), float(row["longitude
48                 "]])))
49         except (KeyError, TypeError, ValueError):
50             continue
51
52     if (len(valid_rows) < 2): return None
53
54     latitudes, longitudes = [r[1] for r in valid_rows], [r[2] for r in
55         valid_rows]
56     altitudes, speeds, satellites = [], [], []
57
58     for r, _, _ in valid_rows:
59         for target, key in [(altitudes, "altitude_m"), (speeds, "speed_mps"), (
60             satellites, "satellites_used")]:
61             try: target.append(float(r[key]))
62             except (KeyError, TypeError, ValueError): pass
63
64     ref_lat, ref_lon = statistics.mean(latitudes), statistics.mean(longitudes)
65     deviations = [distance(ref_lat, ref_lon, lat, lon) for lat, lon in zip(
66         latitudes, longitudes)]
67     changes = [distance(latitudes[i - 1], longitudes[i - 1], latitudes[i],
68         longitudes[i]) for i in range(1, len(latitudes))]
69
70     return {
71         "file": os.path.basename(path), "readings": len(valid_rows),
72         "mean_deviation": statistics.mean(deviations), "std_deviation":
73             statistics.stdev(deviations), "max_deviation": max(deviations),
74         "lat_std": statistics.stdev(latitudes), "lon_std": statistics.stdev(
75             longitudes),
76         "alt_mean": statistics.mean(altitudes) if altitudes else None,
77         "alt_std": statistics.stdev(altitudes) if len(altitudes) > 1 else None,
78         "alt_range": max(altitudes) - min(altitudes) if altitudes else None,
79         "speed_mean": statistics.mean(speeds) if speeds else None, "speed_max":
80             max(speeds) if speeds else None,
81         "sat_mean": statistics.mean(satellites) if satellites else None,
82         "sat_min": min(satellites) if satellites else None, "sat_max": max(
83             satellites) if satellites else None,
84         "change_mean": statistics.mean(changes), "change_max": max(changes),
85         "jumps": sum(c > JUMP_THRESHOLD for c in changes)
86     }

```

Listing 4.2: Core function that produces the metrics in Table 4.1

The GPS-based testing therefore established a baseline for the positional consistency of the GPS component under stationary conditions. Although the receiver produced measurable variation in its reported position, the majority of consecutive observations remained relatively close to one another, demonstrating that the GPS module was able to provide usable positional data through the Raspberry Pi Zero. These results provide a reference against which the GPS observations obtained during the subsequent field test can be compared under flight conditions.

4.4.2 Communication link testing results

4.4.3 Summary of findings

4.5 Conclusion

Bibliography

- [1] Arduino, “Arduino uno rev3,” Arduino Documentation, 2026. [Online]. Available: <https://docs.arduino.cc/hardware/uno-rev3/>
- [2] J. v. Brocke, A. Hevner, and A. Maedche, “Introduction to Design Science Research,” *Progress in IS*, p. 1, 2020.
- [3] B. Brown, “Multi-mode transmitter for high-altitude balloon telemetry,” in *Academic High Altitude Conference*, vol. 2011, no. 1. Iowa State University Digital Press, 2011, pp. 1–6.
- [4] A. Colpaert, E. Vinogradov, and S. Pollin, “Aerial Coverage Analysis of Cellular Systems at LTE and mmWave Frequencies Using 3D City Models,” *Sensors*, pp. 1–13, 2018.
- [5] J. W. Creswell and J. D. Creswell, *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage publications, 2017.
- [6] P. Enge, “The Global Positioning System: Signals, measurements, and performance,” *Int. J. Wirel. Inf. Networks*, 2010.
- [7] M. Y. Feilzer, “Doing mixed methods research pragmatically: Implications for the rediscovery of pragmatism as a research paradigm,” *Journal of Mixed Methods Research*, vol. 4, no. 1, pp. 6–16, 2010.
- [8] T. G. Guzik, “The High Altitude Student Platform (HASP) as a model multi-payload balloon platform,” *2015 IEEE Aerospace Conference*, 2015.
- [9] J. Habermas, *The Theory of Communicative Action: Volume 1*. Beacon Press, 1984.
- [10] Z. Hasirci and I. H. Cavdar, “Propagation modeling dependent on frequency and distance for mobile communications via high altitude platforms (HAPs),” *2012 35th International Conference on Telecommunications and Signal Processing (TSP)*, 2012.
- [11] Independent Communications Authority of South Africa, “National Radio Frequency Plan (NRFP) 2021,” Tech. Rep., 2022, government Gazette No. 46088, 25 March 2022.
- [12] Independent Communications Authority of South Africa (ICASA), “Radio Frequency Spectrum Regulations,” 2011, government Gazette No. 34172, 31 March 2011.
- [13] T. Janssen, A. Koppert, R. Berkvens, and M. Weyn, “A Survey on IoT Positioning Leveraging LPWAN, GNSS, and LEO-PNT,” *IEEE Internet of Things Journal*, pp. 1–34, 2023.
- [14] C. Kivunja and A. B. Kuyini, “Understanding and Applying Research Paradigms in Educational Contexts,” 2017.

- [15] J. Krueger and P. Bergmaier, “Testing Next-Generation Telemetry Using LoRaWAN On High-Altitude Balloons,” *2024 Academic High Altitude Conference*, pp. 1–13, 2024.
- [16] A. Larsson, A. Piotrowski, T. D. Giles, and D. Smart, “Near-earth RF propagation - Path loss and variation with weather,” *2013 International Conference on Radar*, 2013.
- [17] P. Leu, I. Puddu, A. Ranganathan, and S. Čapkun, “I Send, Therefore I Leak: Information Leakage in Low-Power Wide Area Networks,” *arXiv: Cryptography and Security*, pp. 1–13, 2019.
- [18] A. Maxworth, “Worksheet 01 - monopole,” *Antenna Design With HFSS*, 2022. [Online]. Available: https://digitalcommons.usm.maine.edu/oer_maxworth_hfss/10/
- [19] R. Miura, T. Matsuda, F. Ono, T. Matsumura, J. Suzuki, and T. Takamori, “A Study on Spectrum Sharing for the User and Feeder Links of Command and Telemetry Communications for UAVs Using a High Altitude Relay System,” *International Symposium on Wireless Personal Multimedia Communications*, 2021.
- [20] A. Pankine, Z. Li, D. Parsons, M. Purucker, E. Weinstock, W. Wiscombe, and K. Nock, “Stratospheric Satellites for Earth Observations,” 2009.
- [21] Y. S. Park, L. Konge, and A. R. Artino, “The Positivism Paradigm of Research.” *Academic medicine : journal of the Association of American Medical Colleges*, 2020.
- [22] K. Peffers, T. Tuunanen, and B. Niehaves, “Design science research genres: introduction to the special issue on exemplars and criteria for applicable design science research,” *European Journal of Information Systems*, pp. 129–137, 2018.
- [23] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, “A Design Science Research Methodology for Information Systems Research,” vol. 24, no. 3, pp. 45–77, eprint: <https://doi.org/10.2753/MIS0742-1222240302>. [Online]. Available: <https://doi.org/10.2753/MIS0742-1222240302>
- [24] E. D. Poorter, J. Hoebeke, M. Strobbe, I. Moerman, S. Latré, M. Weyn, B. Lannoo, and J. Famaey, “Sub-GHz LPWAN Network Coexistence, Management and Virtualization: An Overview and Open Research Challenges,” *Wireless personal communications*, pp. 1–16, 2017.
- [25] P. Puričar, P. Kovář, J. Mikeš, and J. Kákona, “Telemetry system for research stratospheric balloon,” *2017 International Conference on Applied Electronics (AE)*, 2017.
- [26] Radiometrix Ltd, “Ntx2/nrx2 – uhf narrow band fm tx & rx datasheet,” Radiometrix, 2010, issue 1, 11 February 2010. Accessed September 2026. [Online]. Available: <https://www.radiometrix.com/datasheets/ntx2nrx2.pdf>
- [27] J. Saldaña, “The coding manual for qualitative researchers,” 2021.
- [28] G. Shanks, “Guidelines for Conducting Positivist Case Study Research in Information Systems,” *Australas. J. Inf. Syst.*, 2002.
- [29] P. Techavijit, S. Chivapreecha, P. Sukchalerm, and N. Wongphuangfuthaworn, “Suitable altitude for long-operated communication high altitude balloon with experimental flights,” *2016 8th International Conference on Knowledge and Smart Technology (KST)*, 2016.

- [30] UK High Altitude Society (UKHAS), “Linking an arduino to a radiometrix ntx2b transmitter,” UKHAS Wiki, 2016, accessed September 2026. [Online]. Available: <https://ukhas.org.uk/doku.php?id=guides:linkingarduinotontx2>
- [31] —, “Payload antenna,” UKHAS Wiki, 2016, last modified 7 April 2016. Accessed September 2026. [Online]. Available: https://ukhas.org.uk/doku.php?id=guides:payload_antenna
- [32] M. Uzair, S. Y. Al-Kafrawi, K. M. Al-Janadi, and I. A. Al-Bulushi, “A Low-Cost IoT Based Buildings Management System (BMS) Using Arduino Mega 2560 And Raspberry Pi 4 For Smart Monitoring and Automation,” *International journal of electrical and computer engineering systems*, pp. 1–48, 2022.
- [33] Wang, Zhanchao and Huang, Min and Qian, Lulu and Zhao, Baowei and Wang, Guangming, “High-altitude balloon-based sensor system design and implementation,” *Sensors*, vol. 20, no. 7, pp. 2–3, 2020. [Online]. Available: <https://www.mdpi.com/1424-8220/20/7/2080>
- [34] M. Wing, “Consumer-Grade Global Positioning Systems (GPS) Receiver Performance,” 2008.
- [35] R. O. Wiyagi, T. A. Agus *et al.*, “High Altitude Balloon Payload Design for Atmospheric Observations,” *Journal of Electrical Technology UMY*, vol. 1, no. 1, pp. 50–57, 2017.

Appendix A

Programs

A.1 Ground-based GPS acquisition and logging program

```
1  #!/usr/bin/env python3
2  import gps
3  import csv
4  import json
5  import os
6  import time
7  from datetime import datetime, timezone
8
9  # Only latitude and longitude are required for a valid GPS position
10 REQUIRED_FIELDS = ["lat", "lon"]
11 MIN_MODE = 3 # 2 = 2D fix, 3 = 3D fix
12 LOG_INTERVAL = 5 # seconds
13
14 # Directory containing this Python script
15 SCRIPT_DIR = os.path.dirname(os.path.abspath(__file__))
16
17 # Log directory
18 LOG_DIR = os.path.join(SCRIPT_DIR, "logs")
19
20 # Create logs directory if it does not exist
21 os.makedirs(LOG_DIR, exist_ok=True)
22
23 # Time the program was started
24 START_TIME = datetime.now(timezone.utc)
25
26 # Unique log filename for this execution
27 LOG_FILENAME = (
28     f"gps_log_{START_TIME.strftime('%Y-%m-%d_%H-%M-%S')}.csv"
29 )
30
31 LOG_FILE = os.path.join(LOG_DIR, LOG_FILENAME)
32 # File containing the latest valid GPS information
33 LATEST_FILE = os.path.join(SCRIPT_DIR, "latest_gps.json")
34
35 def is_complete(report):
36     mode = getattr(report, "mode", 0)
```

```

38     if (mode < MIN_MODE):
39         return False, f"mode={mode} (need >= {MIN_MODE})"
40
41     missing = [
42         field for field in REQUIRED_FIELDS
43         if getattr(report, field, None) is None
44     ]
45
46     if (missing):
47         return False, f"missing: {' ', ' '.join(missing)}"
48     return True, None
49
50 def create_log_file():
51     # Create a new CSV file for this program execution.
52     with open(LOG_FILE, "w", newline="") as file:
53         writer = csv.writer(file)
54
55         # Record when this logging session started
56         writer.writerow([
57             f"Logging session started: "
58             f"{START_TIME.isoformat()}"
59         ])
60
61         writer.writerow([
62             "system_time",
63             "satellite_time",
64             "latitude",
65             "longitude",
66             "altitude_m",
67             "speed_mps",
68             "track_deg",
69             "satellites_used"
70         ])
71
72 def log_gps_data(data):
73     # Append a successful GPS fix to the current session log.
74     with open(LOG_FILE, "a", newline="") as file:
75         writer = csv.writer(file)
76
77         writer.writerow([
78             data["system_time"],
79             data["satellite_time"],
80             data["latitude"],
81             data["longitude"],
82             data["altitude_m"],
83             data["speed_mps"],
84             data["track_deg"],
85             data["satellites_used"]
86         ])
87
88 def update_latest_gps(data):
89     # Write the latest valid GPS fix to a JSON file. The temporary file prevents
90     another program from reading a partially-written JSON file.
91     temporary_file = LATEST_FILE + ".tmp"
92     with open(temporary_file, "w") as file:
93         json.dump(data, file, indent=4)

```

```

93     os.replace(temporary_file, LATEST_FILE)
94
95 def main():
96     create_log_file()
97     session = gps.gps(mode=gps.WATCH_ENABLE)
98     print("GPS monitoring started.")
99     print(f"Logging to: {LOG_FILE}")
100    print(f"Latest GPS data: {LATEST_FILE}")
101    print(f"Valid GPS fixes are logged every {LOG_INTERVAL} seconds.")
102    print("Press Ctrl+C to stop.\n")
103    last_log_time = 0
104
105    while (True):
106        try:
107            report = session.next()
108        except StopIteration:
109            continue
110        except KeyError:
111            continue
112
113        if (report.get("class") != "TPV"):
114            continue
115
116        ok, reason = is_complete(report)
117
118        if (not ok):
119            continue
120
121        sats = getattr(session, "satellites_used", 0)
122
123        # Optional GPS values are allowed to be unavailable
124        altitude = getattr(report, "alt", None)
125        speed = getattr(report, "speed", None)
126        track = getattr(report, "track", None)
127
128        # System time from the Raspberry Pi
129        system_time = datetime.now(timezone.utc).isoformat()
130
131        # GPS time received from the satellites through the GPS receiver
132        satellite_time = getattr(report, "time", None)
133
134        latest_valid_fix = {
135            "system_time": system_time,
136            "satellite_time": satellite_time,
137            "latitude": report.lat,
138            "longitude": report.lon,
139            "altitude_m": altitude,
140            "speed_mps": speed,
141            "track_deg": track,
142            "satellites_used": sats
143        }
144
145        # ALWAYS update the latest GPS information.
146        # This means geofencing/transmission can access
147        # the newest valid position independently of
148        # the 5-second research logging interval.

```

```

149         update_latest_gps(latest_valid_fix)
150
151         current_time = time.monotonic()
152
153         # Log one valid GPS measurement every 5 seconds
154         if (current_time - last_log_time >= LOG_INTERVAL):
155             log_gps_data(latest_valid_fix)
156             print(
157                 f"SYSTEM TIME: {system_time} | "
158                 f"SATELLITE TIME: "
159                 f"{satellite_time if satellite_time is not None else 'N/A'}"
160             )
161             print(
162                 f"VALID FIX LOGGED "
163                 f"lat={report.lat:.6f} "
164                 f"lon={report.lon:.6f} "
165                 f"alt={altitude if altitude is not None else 'N/A'}m "
166                 f"speed={speed if speed is not None else 'N/A'}m/s "
167                 f"track={track if track is not None else 'N/A'}deg "
168                 f"sats_used={sats}"
169             )
170             last_log_time = current_time
171
172 if (__name__ == "__main__"):
173     try:
174         main()
175     except KeyboardInterrupt:
176         print("\nGPS monitoring stopped.")

```

A.2 GPS data analysis program

```

1  #!/usr/bin/env python3
2
3  import csv, math, os, statistics
4  from datetime import datetime
5
6  # CONFIGURATION
7  BASE_DIR = os.path.dirname(os.path.abspath(__file__))
8  LOG_DIR = os.path.join(BASE_DIR, "logs")
9  OUTPUT_FILE = os.path.join(BASE_DIR, "log_analysis.txt")
10 JUMP_THRESHOLD = 5.0 # metres
11
12 # HELPER FUNCTIONS
13 def distance(lat1, lon1, lat2, lon2):
14     #Calculate distance between two GPS coordinates in metres.
15     R = 6371000
16     p1, p2 = math.radians(lat1), math.radians(lat2)
17     dp, dl = math.radians(lat2 - lat1), math.radians(lon2 - lon1)
18     a = min(1.0, max(0.0, math.sin(dp / 2) ** 2 + math.cos(p1) * math.cos(p2) *
19                 math.sin(dl / 2) ** 2))
19     return R * 2 * math.atan2(math.sqrt(a), math.sqrt(1 - a))
20
21 def fmt(value, unit=""):
22     #Format numerical values for the report.

```

```

23     return "N/A" if value is None else f"{value:.3f}{unit}" if isinstance(value,
24         float) else f"{value}{unit}"
25
26 def make_table(headers, rows):
27     #Create a simple text table.
28     if (not rows): return []
29     widths = [max(len(str(headers[i])), max(len(str(row[i])) for row in rows))
30         for i in range(len(headers))]
31     sep = "+-" + "-+-".join("-" * w for w in widths) + "-+"
32     header = "| " + " | ".join(str(headers[i]).ljust(widths[i]) for i in range(
33         len(headers))) + " |"
34     return [sep, header, sep] + ["| " + " | ".join(str(row[i]).ljust(widths[i])
35         for i in range(len(headers))) + " |" for row in rows] + [sep]
36
37 # GPS LOG ANALYSIS
38 def analyse_file(path):
39     #Analyse one GPS CSV file.
40     with open(path, newline="", encoding="utf-8") as f:
41         f.readline() # Skip metadata header line
42         reader = csv.DictReader(f)
43         if (not {"latitude", "longitude", "altitude_m", "speed_mps", "
44             satellites_used"}.issubset(reader.fieldnames or [])):
45             return None
46         rows = list(reader)
47
48     valid_rows = []
49     for row in rows:
50         try:
51             valid_rows.append((row, float(row["latitude"]), float(row["longitude
52                 "]))
53         except (KeyError, TypeError, ValueError):
54             continue
55
56     if (len(valid_rows) < 2): return None
57
58     latitudes, longitudes = [r[1] for r in valid_rows], [r[2] for r in
59         valid_rows]
60     altitudes, speeds, satellites = [], [], []
61
62     for r, _, _ in valid_rows:
63         for target, key in [(altitudes, "altitude_m"), (speeds, "speed_mps"), (
64             satellites, "satellites_used")]:
65             try: target.append(float(r[key]))
66             except (KeyError, TypeError, ValueError): pass
67
68     ref_lat, ref_lon = statistics.mean(latitudes), statistics.mean(longitudes)
69     deviations = [distance(ref_lat, ref_lon, lat, lon) for lat, lon in zip(
70         latitudes, longitudes)]
71     changes = [distance(latitudes[i - 1], longitudes[i - 1], latitudes[i],
72         longitudes[i]) for i in range(1, len(latitudes))]
73
74     return {
75         "file": os.path.basename(path), "readings": len(valid_rows),
76         "mean_deviation": statistics.mean(deviations), "std_deviation":
77             statistics.stdev(deviations), "max_deviation": max(deviations),

```

```

67     "lat_std": statistics.stdev(latitudes), "lon_std": statistics.stdev(
68         longitudes),
69     "alt_mean": statistics.mean(altitudes) if altitudes else None,
70     "alt_std": statistics.stdev(altitudes) if len(altitudes) > 1 else None,
71     "alt_range": max(altitudes) - min(altitudes) if altitudes else None,
72     "speed_mean": statistics.mean(speeds) if speeds else None, "speed_max":
73         max(speeds) if speeds else None,
74     "sat_mean": statistics.mean(satellites) if satellites else None,
75     "sat_min": min(satellites) if satellites else None, "sat_max": max(
76         satellites) if satellites else None,
77     "change_mean": statistics.mean(changes), "change_max": max(changes),
78     "jumps": sum(c > JUMP_THRESHOLD for c in changes)
79 }
80
81 # MAIN
82 def main():
83     print(f"\n{'=' * 78}\n\n                                GPS LOG ANALYSIS\n{'=' * 78}"
84           )
85     if (not os.path.exists(LOG_DIR)):
86         os.makedirs(LOG_DIR)
87         print(f"\nLog directory created at: {LOG_DIR}\nPlease place .csv log
88             files in this directory and re-run.")
89     return
90
91     files = sorted(fn for fn in os.listdir(LOG_DIR) if fn.lower().endswith(".csv
92         "))
93     total_files, results, empty_files = len(files), [], []
94
95     for filename in files:
96         path = os.path.join(LOG_DIR, filename)
97         try:
98             res = analyse_file(path)
99             if (res is None or res["readings"] < 2):
100                 empty_files.append(filename)
101                 print(f"    [EMPTY]      {filename}")
102             else:
103                 results.append(res)
104                 print(f"    [ANALYSED] {filename:<42}{res['readings']:>4} readings
105                     ")
106         except Exception as error:
107             empty_files.append(filename)
108             print(f"    [ERROR]      {filename} ({error})")
109
110     total_readings = sum(r["readings"] for r in results)
111
112     # BUILD REPORT
113     report = [
114         f"\n{'=' * 78}\n\n                                GPS LOG ANALYSIS\n{'=' * 78}"
115         f"Generated: {datetime.now().astimezone().isoformat()}\n\n"
116         f"DETAILED MEASUREMENTS\n{'-' * 78}"
117     ]
118
119     if (results):
120         for r in results:
121             report.append(
122                 f"\nFILE: {r['file']}\nReadings: {r['readings']}\n\n"

```

```

116         f" Positional deviation: mean={fmt(r['mean_deviation'], ' m')},
           std={fmt(r['std_deviation'], ' m')}, max={fmt(r['
117         max_deviation'], ' m')}\n"
118         f" Position variation: latitude std={fmt(r['lat_std'])},
           longitude std={fmt(r['lon_std'])}\n"
119         f" Altitude: mean={fmt(r['alt_mean'], ' m')}, std={fmt(r['
           alt_std'], ' m')}, range={fmt(r['alt_range'], ' m')}\n"
120         f" Speed: mean={fmt(r['speed_mean'], ' m/s')}, max={fmt(r['
           speed_max'], ' m/s')}\n"
121         f" Satellites: mean={fmt(r['sat_mean'])}, min={fmt(r['sat_min
           '])}, max={fmt(r['sat_max'])}\n"
122         f" Consecutive position changes: mean={fmt(r['change_mean'], '
           m')}, max={fmt(r['change_max'], ' m')}, jumps > {
           JUMP_THRESHOLD:.0f} m={r['jumps']}"
123     )
124 else:
125     report.append("\nNo usable GPS logs were found.")
126
127 # INDIVIDUAL LOG RESULTS
128 report.append(f"\n\nINDIVIDUAL LOG RESULTS\n{'-' * 78}\n")
129 if (results):
130     headers = ["Log file", "Readings", "Mean Dev.", "Max Dev.", "Mean Change",
131               "Max Change", "Jumps"]
132     rows = [[r["file"], r["readings"], fmt(r["mean_deviation"], " m"), fmt(r[
133               "max_deviation"], " m"), fmt(r["change_mean"], " m"), fmt(r["
134               change_max"], " m"), r["jumps"]] for r in results]
135     report.extend(make_table(headers, rows))
136 else:
137     report.append("No usable GPS logs were found.")
138
139 # OVERALL SUMMARY
140 report.append(f"\n\n{'=' * 78}\n\n                                OVERALL SUMMARY\n
141               {'=' * 78}\n")
142 report.extend(make_table(["Measurement", "Value"], [
143     ["Total CSV files read", total_files],
144     ["Files containing GPS values", len(results)],
145     ["Empty CSV files", len(empty_files)],
146     ["Total GPS readings", total_readings]
147 ]))
148
149 if (results):
150     m_devs = [r["mean_deviation"] for r in results]
151     chgs = [r["change_mean"] for r in results if r["change_mean"] is not
152             None]
153     metrics = [
154         ["Average positional deviation", f"{statistics.mean(m_devs):.3f} m"
155         ],
156         ["Largest positional deviation", f"{max(r['max_deviation'] for r in
157         results):.3f} m"],
158         ["Average consecutive position change", f"{statistics.mean(chgs):.3f
159         } m" if chgs else "N/A"],
160         [f"Total jumps > {JUMP_THRESHOLD:.0f} m", sum(r["jumps"] for r in
161         results)]
162     ]
163     report.append("\n")
164     report.extend(make_table(["GPS Metric", "Result"], metrics))

```

```

155
156 # INTERPRETATION
157 report.append(
158     f"\n\nINTERPRETATION\n{'-' * 78}\n\n"
159     "Each CSV is analysed independently using its own mean latitude/
160     longitude as the reference position.\n\n"
161     "Therefore, positional deviation represents GPS consistency/
162     repeatability rather than absolute accuracy against a surveyed
163     reference coordinate.\n\n"
164     "For a stationary receiver, speed should ideally remain close to 0 m/s.
165     Consecutive position changes therefore indicate apparent GPS
166     movement or measurement variation.\n\n"
167     f"A position change greater than {JUMP_THRESHOLD:.0f} m is classified as
168     a jump.\n\n"
169     "Track is not included because it is not particularly meaningful when
170     the GPS receiver is stationary."
171 )
172
173 # SAVE REPORT
174 report_text = "\n".join(report)
175 print(f"\n{report_text}")
176 with open(OUTPUT_FILE, "w", encoding="utf-8") as f: f.write(report_text)
177 print(f"\n{'=' * 78}\nReport saved to: {OUTPUT_FILE}\n{'=' * 78}")
178
179 # EMPTY FILE CLEANUP
180 if (empty_files):
181     print(f"\n{'-' * 78}\nEMPTY CSV CLEANUP\n{'-' * 78}\n\nTotal files read:
182         {total_files}\nFiles with GPS values: {len(results)}\nEmpty
183         files: {len(empty_files)}\n\nA CSV with fewer than 2
184         valid GPS readings is considered empty.\n")
185     if (input("Delete the empty CSV files? [y/N]: ").strip().lower() in ("y"
186         , "yes")):
187         deleted = 0
188         for fn in empty_files:
189             try:
190                 os.remove(os.path.join(LOG_DIR, fn))
191                 deleted += 1
192                 print(f" Deleted: {fn}")
193             except OSError as err:
194                 print(f" Could not delete {fn}: {err}")
195         print(f"\nDeleted {deleted} of {len(empty_files)} empty CSV file(s).
196             ")
197     else:
198         print("\nNo files were deleted.")
199
200 if (__name__ == "__main__"):
201     main()

```

Appendix B

B.1 Code of Conduct

CODE OF CONDUCT FOR RESEARCHERS

This code of conduct is applicable to all NWU researchers.

As a researcher of the North-West University (NWU), I subscribe to the rules of the NWU Institutional Research Ethics Regulatory Committee (IRERC), all applicable policies of the NWU as well as all national and international laws and regulations applicable to my field of study. Furthermore, I commit myself to abide by the ethical principles and responsibilities as set out in the Singapore statement on Research Integrity (22 September 2010), in any and all research endeavours that I undertake as a researcher of the NWU.

The four major principles of research integrity to which I will adhere and that will guide my research are:

- Honesty in all aspects of research;
- Accountability in the conduct of research;
- Professional courtesy and fairness in working with others;
- Good stewardship of research on behalf of others.

Consequently I will also adhere to the following ethical responsibilities:

1. I will take responsibility for the originality and trustworthiness of my research.
2. I will stay abreast of and adhere to all institutional, national, and international laws, regulations, and policies applicable and related to my research.
3. I will at all times employ appropriate research methods, base my conclusions on critical analysis of the evidence and report my findings and interpretations fully and objectively.
4. I will keep clear and accurate records of all research that I have conducted in a manner that will allow verification and replication of my work by others, if applicable.
5. I will, where applicable, share my data and findings openly and promptly, in line with external funding rules. This will be done as soon as possible after I have had an opportunity to establish priority and ownership claims.
6. I will take responsibility for my own contributions to publications, funding applications, reports and other representations of my research. I will also and only include authors who meet valid authorship criteria.
7. I will acknowledge the names and roles of those who made significant contributions to my research in publications, including writers, funders, sponsors, and others, but do not meet authorship criteria.
8. In my peer reviews, I will provide fair, prompt and rigorous evaluations and I will respect confidentiality when I review others' work.
9. I will disclose all conflicts of interest (financial and other) that could compromise the trustworthiness of my work in research proposals, publications, public communications, and in review activities.
10. When I publically address a community in the spirit of academic freedom, I will in all stages base my professional comments on research findings (if applicable) and my expertise. I will distinguish between professional comments and opinions based on personal views.
11. Should any irresponsible research practices and/or research misconduct become known to me or brought under my attention, I will report such irresponsible research activities to the appropriate authorities.
12. I will respond to irresponsible research practices or conduct, by taking prompt actions as set out in the procedures of the university. I will also protect those who report misconduct in good faith, to the best of my abilities.
13. I will endeavour to create and sustain an environment that encourage research integrity through education of students, research teams and peers, as well as abide by policies, and reasonable standards for advancement.
14. I will at all times weigh societal benefits against the risks inherent in my work.

Name: Denzel Verster

Signature:



Date: 07/04/2026

B.2 Student Study Leader Agreement

ITRI671 Research Project Agreement

This document will serve as a contract between the student and the supervisor.

Name of student: Denzel Verster

Student number: 45036349

Name of supervisor: Professor Gunther Drevin

To ensure the success of your project, the following agreement is made between the supervisor and the student:

1.	Identifying the keywords and possible sources is the responsibility of the student and not of the supervisor.
2.	For any literature study, the student must make use of books, articles, conference papers and internet sources. (NO Wikipedia or questionable websites)
3.	It is the responsibility of the student to identify a primary objective and secondary objectives for the study. The supervisor will give guidance as to whether the objectives are achievable.
4.	Improvements on feedback received from the supervisor, e.g. formulation of statements, language use, referencing and technical finishing of the document are the responsibility of the student.
5.	Plagiarism is reviewed per chapter. A warning will be issued if plagiarism is detected. A second warning will result in disciplinary steps being taken. Adhere to the NWU guidelines regarding AI use.
6.	Keep to the schedule agreed upon and submit work on time. Late/No submission will result in a warning.
7.	Feedback from the supervisor will be given within an agreed upon timeframe after submission.
8.	Attend all meetings scheduled, be on time and be prepared. Mutual respect will be given to the time of the supervisor and the student.
9.	If any uncertainty or problem arises, the student will seek clarity from the supervisor immediately.
10.	The content of all the documentation and the empirical analysis must be the student's own work. The supervisor is not a source but should rather be seen as a guide (sounding board).
11.	The supervisor may at any time request the student to make use of the services of a language editor. The supervisor will not be responsible for language editing. Please ensure that your work is of a high standard in this regard.

Signature of student:  Date: 07/04/2026

Signature of supervisor: _____ Date: _____

*If the student or supervisor is not satisfied with the effort of the other party –
please contact the module facilitator immediately to resolve the issue.*

B.3 Declaration of Integrity

Plagiarism, Artificial Intelligence and Academic Integrity

- When you submit an assignment, it should always be your own individual work unless clearly indicated as a group activity.
- Copying text from other learners or sources without giving credit by citing the source and adding the source to your reference list is considered plagiarism.
- When you use information from a source, you should paraphrase the text. This means reformulating the information in your own words to explain what you have read. The assignment aims not to reproduce existing material but to ascertain whether you can integrate texts, add your own interpretation and/or critique the texts, and offer a creative solution to existing problems.
- It is your responsibility to familiarise yourself with the [NWU Academic Integrity Policy](#) and not make yourself guilty of [plagiarism](#).
- It is generally assumed that using Artificial Intelligence (AI) tools such as ChatGPT is not allowed unless clearly stated. You should note that the NWU uses Turnitin to detect plagiarism and/or the use of Artificial Intelligence.
- You are responsible for confirming your lecturers' stance on using AI and familiarising yourself with the [NWU Guidelines on the Responsible and Ethical use of Artificial Intelligence](#).
- Remember that language editing tools and/or paraphrasing tools are considered a form of AI and might be flagged by Turnitin. Always save a copy of your original, unedited work before you submit it to any of these tools.
- Plagiarism or the unethical use of AI is a serious offence. Disciplinary steps may include a significant deduction of marks and/or being reported on the AIITSA system. For more information, visit the NWU page on [Academic Integrity](#).
- It is also unethical and unacceptable to do someone else's work, lend it to someone else, or make it available to be copied.

DECLARATION

Name: Denzel Verster

Student number: 45036349

Module: ITRI671

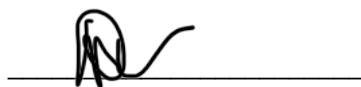
Supervisor: Professor Gunther Drevin

Title of the Research Project:

Communication and Tracking System for a Stratospheric Balloon

I hereby declare that:

1. The above research report is my own work and that all the sources that I have used or quoted have been indicated and acknowledged by means of complete references.
2. I submitted the research report to originality checking software and that it falls within the accepted requirements for originality (under supervisor of the supervisor).
3. No data presented in this work is fabricated or falsified.
4. I adhere to all the Plagiarism, Artificial Intelligence and Academic Integrity guidelines above.



Signature of Student

Date 07/04/2026

B.4 Ethics Form

ITRI 671 Research ethics form: Honours project

Name: Denzel Verster

Title of project: Communication and Tracking System for a Stratospheric Balloon

Supervisor: Prof. Gunther Drevin

Co-supervisor: Mr. J. Heystek Grobler

Co-supervisor: Dr. Rudi Serfontein

Starting and end dates of project: 11 February 2026– 19 November 2026

Please discuss the ethical clearance with your supervisor by using the **General principles and guidelines document on eFundi..**

Indicate the ethics level.

Level for this project	1	2 ☒	3
------------------------	---	---------------------------------------	---

1. Have you read the information available related to research ethics (Chapter 5 of Researching Information Systems and Computing; BJ Oates and Chapter 13 of Writing for computer science, J Zobel; Manual for post graduate studies, available on efundi)?	Yes ☒	No
2. Do you make use of people as source of data in your project (for example the completion of questionnaires or evaluation of products)? In what way? This study does not involve human participants as a source of data. All data will be generated through system testing, including ground-based experiments and high-altitude balloon launches.	Yes	No ☒